Measure Equivalence as a Guide for Learning SPDE Priors from Replicated Fields

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Abstract

Learning a Gaussian prior from replicated spatial fields requires deciding not only how much flexibility to add, but where that flexibility belongs. We study whether a population covariance discrepancy can be represented as a correction of a reference Gaussian field or instead requires changing the differential operator that defines the reference. Using a leaky-aquifer SPDE, we compare finite-dimensional source variability, spatially varying leakage, and spatially varying conductivity. Gaussian-measure equivalence theory predicts that the first two remain within the reference measure class, whereas a principal-coefficient change does not. We translate this distinction into a refinement diagnostic based on reference-whitened replicated fields. For a conductivity lens, the diagonal refinement statistic $\widehat{h}_N$ computed from 200 fields tracks the finite-grid oracle sequence toward the nonzero constant predicted from the principal symbol. Wrong structural classes fail systematically: source variability drives a stationary range estimate from the generating value $\kappa_0 = 6$ to 3×10^{-5}, while correction-only models incur a cost growing linearly with resolution on principal-part variation. A combined model that jointly adapts the principal operator and learns finite-rank source corrections returns $\widehat{\kappa} = 6.08$ on the same source-only truth and recovers the conductivity field and source subspace in the mixed case. Whether to correct a Gaussian reference or change its operator is a question that replicated data can answer, and getting that decision wrong can produce a confident, physically plausible, but structurally incorrect model.

1 Introduction

Gaussian random fields are widely used as probabilistic models for spatially varying quantities, and stochastic partial differential equation (SPDE) representations provide a useful connection between covariance structure and local differential operators (Lindgren et al., 2011, 2022). When replicated realizations of a spatial field are available, they provide information about population-level covariance structure beyond that contained in a single realization. Replicated fields have therefore been used to estimate nonstationary spatial dependence and to separate persistent population structure from realization-specific variation (Ingebrigtsen et al., 2015; Fuglstad et al., 2015). This leads to the question studied here: can replicated fields reveal whether a covariance discrepancy can be represented as a correction of a chosen reference model, or whether the operator defining that reference must itself change?

A common response to mismatch is to introduce flexibility around a reference model. One class of methods augments a simpler covariance model with low-dimensional or low-rank structure, as in fixed-rank kriging, predictive processes, and full-scale covariance approximations (Cressie and Johannesson, 2008; Banerjee et al., 2008; Sang and Huang, 2012). Predictive nonstationary constructions such as LatticeKrig, nearest-neighbor Gaussian processes (NNGP), and fused Gaussian processes likewise operate on the correction

side of a fixed reference and are therefore subject to the same reference-adequacy diagnostic considered here. A second class modifies the covariance-generating operator, for example by allowing spatially varying range, anisotropy, or coefficients in an SPDE representation (Fuglstad et al., 2015; Lindgren et al., 2022). In Bayesian formulations, penalized-complexity (PC) priors provide a principled way to shrink a flexible model component toward a simpler base model (Simpson et al., 2017; Fuglstad et al., 2019). In each case the reference is taken as given; whether it is the right reference for the observed population is not asked.

For Gaussian fields in function space, departures from a reference can have different mathematical character. By the Feldman–Hájek dichotomy, two Gaussian measures are either equivalent or mutually singular. For generalized Whittle–Matérn fields generated by elliptic operators, Bolin and Kirchner (2023) give conditions under which changes of the operator preserve or destroy equivalence. Here the *principal part* is the highest-order differential term, for example $-\nabla\cdot(K(x)\nabla)$ in a second-order elliptic operator, while κ^2 is a zeroth-order term. In the dimensions considered here, changes to certain lower-order terms can remain within the same Gaussian measure class, whereas a change in the principal coefficient does not preserve equivalence: equality of the principal coefficient is required, with even a constant rescaling acting as a covariance-amplitude change. Finite-dimensional source perturbations can also remain within the reference measure class when the induced field directions satisfy the Cameron–Martin requirement. In the setting studied below, smooth source profiles satisfy this requirement, whereas idealized point sources do not. Thus source, lower-order, and principal-part perturbations can all appear as covariance changes on a finite mesh while having different continuum behavior.

This distinction is directly relevant to methods that aim to learn probability measures independently of discretization. Function-space diffusion models introduce Gaussian reference or perturbing measures precisely so that the learned construction remains meaningful as resolution changes. Lim et al. (2025) formulate score-based diffusion in function space under absolute-continuity and Cameron–Martin conditions relative to a Gaussian perturbing measure, while Pidstrigach et al. (2023) show that discretize-first diffusion constructions can deteriorate under mesh refinement and instead develop an infinite-dimensional formulation with dimension-independent guarantees. The inverse-problem setting provides a useful safe-case contrast: with finite-dimensional observations, the Gaussian posterior differs from its prior only through a finite-dimensional information update, and Carere and Lie (2026) characterize low-rank posterior approximations that remain equivalent to both prior and posterior. Population-prior learning has no analogous guarantee that the observed population remains in the measure class of the chosen reference.

The statistical problem is therefore different from comparing two specified continuum models. In prior learning, the reference model and a finite collection of replicated realizations are observed, while the mechanism producing the covariance discrepancy is unknown. Every pair of nonsingular Gaussian distributions on a fixed finite-dimensional discretization is mutually absolutely continuous, so a single mesh does not reveal the continuum distinction directly. Classical fixed-domain results add a second difficulty: covariance parameters can be weakly identified even when prediction remains accurate, and misspecified range parameters can produce asymptotically similar interpolation behavior (Stein, 1999; Zhang, 2004; Kaufman and Shaby, 2013). In particular, the range parameter provides a weakly constrained low-frequency direction through which a stationary fit can absorb localized source variability. A finite-resolution model can therefore be statistically good while assigning variability to the wrong structural component.

The broader problem of mechanism misattribution is already recognized in spatial statistics. Fuglstad et al. (2015) show that apparent covariance nonstationarity can arise from inadequately modeled mean structure: missing spatial covariates can be absorbed by the covariance model, creating spurious long-range dependence or apparent global nonstationarity. Related work with replicated spatial fields uses repeated realizations to help separate competing sources of spatial variation (Ingebrigtsen et al., 2015). At the SPDE level, Cui and He (2027) derive identifiability conditions for drift and diffusion components from trajectory data using mean and covariance structure; their problem concerns dynamical SPDE discovery rather than covariance attribution across replicated equilibrium fields. Other nearby work fixes one side of the distinction. Bolin

(2026) places known covariates into the forcing of an SPDE to separate pointwise and operator-mediated mean effects, while Petersson and Schroers (2025) develops estimation and goodness-of-fit tests for the driving-noise covariance with the differential operator treated as given. The question considered here is instead whether a covariance discrepancy across replicated fields should be assigned to the source covariance or to the spatial operator generating the field, and how those alternatives behave under mesh refinement.

This distinction also matters for reference-based prior construction. PC priors formalize model complexity as departure from a chosen base model (Simpson et al., 2017). For Matérn fields, the construction is parameterized so that the relevant distance from the base remains meaningful (Fuglstad et al., 2019). More recent work on anisotropic SPDEs shows that principal-operator changes can produce mutually singular Gaussian measures, so alternative complexity measures are required when the usual Kullback–Leibler construction is unavailable (Llamazares-Elias et al., 2026). These results motivate a question that precedes regularization around a base model: is the chosen reference operator adequate for the observed population? A within-class departure can be represented and regularized relative to that reference, whereas principal-part variation requires adaptation of the operator defining the reference covariance. On a fixed mesh, forcing a principal-part change into a reference-based correction can still produce a finite KL discrepancy, but that discrepancy can grow with resolution, making the implied complexity itself discretization dependent.

We study this question using a leaky-aquifer SPDE as a controlled physical setting because uncertain pumping, spatially varying leakage, and heterogeneous conductivity enter respectively through the source, a lower-order term, and the principal part of the operator. Let $\Sigma_{0,N}$ denote the covariance of the nominal reference model on a discretization with N degrees of freedom, and let $\Sigma_{1,N}$ denote the population covariance of the field-generating process on the same discretization. We define the relative covariance departure in reference-whitened coordinates by

$$T_N = \Sigma_{0,N}^{-1/2} \Sigma_{1,N} \Sigma_{0,N}^{-1/2} - I. \tag{1}$$

The population covariance $\Sigma_{1,N}$ is unknown in practice. Given replicated fields $u^{(1)}, \ldots, u^{(n)}$, it is replaced by the sample covariance $\widehat{\Sigma}_{1,N}$, giving

$$\widehat{T}_N = \Sigma_{0,N}^{-1/2} \widehat{\Sigma}_{1,N} \Sigma_{0,N}^{-1/2} - I. \tag{2}$$

The continuum theory characterizes the limiting structure of T_N; the statistical problem is to infer that structure from $\widehat{T}_N$ at finite sample size and finite resolution. We use its spectral and refinement behavior to distinguish source-like covariance corrections from persistent principal-part discrepancy and to study the failures produced by assigning the discrepancy to the wrong structural component.

The resulting modeling principle is to adapt the reference operator when the replicated population exhibits principal-part variation and to represent finite-dimensional source variability as a correction relative to the adapted reference. We implement this principle in a combined model that learns a spatially varying principal coefficient together with a low-rank covariance correction. The operator component and correction component are selected separately from the data, so source-like and operator-level variability need not imitate one another.

The contributions of this work are fourfold. First, we use established Gaussian-measure equivalence results to formulate a statistical prior-learning problem that distinguishes source, lower-order, and principal-part departures through their behavior under discretization refinement (Section 5.1). Second, we develop a sample-based whitening and refinement diagnostic for persistent principal-part mismatch and evaluate how its finite-sample behavior changes when the departure is not separated from the sample covariance bulk as an isolated eigenvalue (Section 5.2). Third, we characterize asymmetric mechanism-misattribution failures when source variability is represented through the operator or principal-part variability is represented through a correction-only model; we also show that the KL cost of the latter can grow with resolution, linking structural misspecification to discretization-dependent reference-based complexity (Section 5.3). Fourth, we introduce a combined model that learns the principal operator and finite-dimensional source corrections jointly, allowing each form of variability to be represented by the component suited to its structure (Section 5.4).

2 Problem formulation and theoretical basis

This section defines the population-level distinction that will later be inferred from finite replicated data. The aim is to identify what separates a covariance correction that remains in the Gaussian measure class of a reference field from a departure that changes the covariance-generating operator itself.

2.1 Reference field and population-learning problem

Let $\mathcal{D} \subset \mathbb{R}^2$ be a bounded domain and consider the elliptic SPDE

$$A_0 u = \mathcal{W}, \qquad A_0 = \kappa_0^2 I - \nabla \cdot (K_0 \nabla), \tag{3}$$

with homogeneous Dirichlet boundary conditions. Here $\mathcal{W}$ denotes spatial Gaussian white noise, κ_0^2 is a zeroth-order coefficient, and K_0 is the principal coefficient. The nominal reference used below has $K_0 \equiv 1$. At the continuum level, we take $\mathcal{W}$ to have identity covariance in the formal white-noise coordinates. Under the present boundary conditions A_0 is positive and self-adjoint, so the centered reference field has law

$$\mu_0 = \mathcal{N}(0, \Sigma_0), \qquad \Sigma_0 = A_0^{-1} A_0^{-1} = A_0^{-2}. \tag{4}$$

Equation (4) is a continuum operator identity. Its discrete form depends on the representation of white noise. For the finite-difference scheme used in the numerical study, nodal white noise has covariance $Q_W = h^{-2} I_N$, giving $\Sigma_{0,N} = \sigma^2 h^{-2} A_{0,N}^{-1} A_{0,N}^{-T}$. The complete discretization is specified in Section 4.

The value $\kappa_0 = 0$ remains admissible on a bounded domain with homogeneous Dirichlet conditions. In that case $A_0 = -\Delta_D$, whose smallest eigenvalue is strictly positive, so the inverse and the covariance remain well defined. This observation will matter when fitting stationary reference models, because decreasing κ corresponds to increasing the spatial range without making the bounded-domain covariance singular at $\kappa = 0$.

We assume that replicated fields

$$u^{(1)}, \ldots, u^{(n)} \overset{\text{iid}}{\sim} \mu_1 = \mathcal{N}(0, \Sigma_1) \tag{5}$$

are generated from an unknown population law. The prior-learning problem is to determine how Σ_1 differs structurally from the known reference covariance Σ_0. In particular, the population departure may arise through the stochastic source, through a lower-order term in the operator, or through the principal coefficient itself.

2.2 Relative covariance in reference-whitened coordinates

The comparison becomes transparent after removing the covariance already represented by the reference. Define

$$T = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} - I. \tag{6}$$

If

$$z = \Sigma_0^{-1/2} u, \tag{7}$$

then the reference field satisfies $\mathrm{Cov}_{\mu_0}(z) = I$, while under the population law

$$\mathrm{Cov}_{\mu_1}(z) = I + T. \tag{8}$$

Thus T is the covariance structure that remains after whitening by the reference model.

On a discretization with N degrees of freedom, the corresponding matrix is

$$T_N = \Sigma_{0,N}^{-1/2} \Sigma_{1,N} \Sigma_{0,N}^{-1/2} - I_N. \tag{9}$$

Its eigenvalues $t_{j,N}$ quantify relative covariance departures mode by mode. The accumulated squared relative departure is

$$D_N := \|T_N\|_{\mathrm{HS}}^2 = \sum_{j=1}^{N} t_{j,N}^2. \tag{10}$$

The refinement behavior of D_N will provide the bridge between continuum equivalence and the finite-dimensional statistics used later.

2.3 Gaussian equivalence and the Hilbert–Schmidt condition

For two Gaussian measures on an infinite-dimensional Hilbert space, the Feldman–Hájek theorem gives an equivalence-or-singularity dichotomy. In one standard formulation, equivalence requires: (i) equality of the Cameron–Martin spaces, (ii) the mean difference to lie in that common space, and (iii) an appropriate relative covariance operator to be Hilbert–Schmidt (Da Prato and Zabczyk, 2014, Thm. 2.25). All fields considered here are centered, so the mean condition is automatic. For the elliptic families below, the common-domain and Cameron–Martin conditions are supplied by the corresponding operator results; the condition that distinguishes the cases of interest is therefore the Hilbert–Schmidt behavior of the relative covariance.

Under these conditions,

$$T \in \mathcal{S}_2 \qquad \Longleftrightarrow \qquad \|T\|_{\mathrm{HS}}^2 = \sum_{j=1}^{\infty} t_j^2 < \infty, \tag{11}$$

where $\mathcal{S}_2$ denotes the Hilbert–Schmidt class. We use this condition to distinguish departures whose cumulative relative effect decays sufficiently rapidly across fine scales from departures that persist through the high-frequency spectrum.

2.4 Three forms of population departure

The leaky-aquifer example provides three physically distinct perturbations of (3): uncertainty in pumping acts through the source, spatially varying leakage changes the zeroth-order term, and heterogeneous conductivity changes the principal part.

2.4.1 Source variability

Let the source contain r random smooth profiles,

$$A_0 u = \mathcal{W} + G\xi, \qquad G\xi = \sum_{k=1}^{r} \xi_k g_k, \qquad \xi \sim \mathcal{N}(0, C). \tag{12}$$

The resulting covariance is

$$\Sigma_1 = A_0^{-1} \left(I + GCG^*\right) A_0^{-1}. \tag{13}$$

Using the continuum identity $\Sigma_0^{1/2} = A_0^{-1}$, whitening gives

$$T = GCG^*. \tag{14}$$

Hence $\mathrm{rank}(T) \le r$ and T is Hilbert–Schmidt.

For $\mu_0 = \mathcal{N}(0, A_0^{-2})$, the Cameron–Martin space is $\mathrm{Ran}(A_0^{-1})$. The induced field direction $A_0^{-1} g_k$ therefore belongs to the Cameron–Martin space exactly when the forcing profile satisfies $g_k \in L^2(\mathcal{D})$. Smooth well profiles satisfy this condition, whereas idealized delta sources do not. The smooth-source assumption is therefore load-bearing for the finite-rank in-class construction.

2.4.2 Spatially varying leakage

Let the principal coefficient remain fixed while the zeroth-order coefficient changes,

$$A_1 = -\nabla\cdot(K_0\nabla) + \kappa^2(x). \tag{15}$$

For the smooth setting covered by Bolin and Kirchner (2023, Thm. 4.11 and Table 1), changes of this lower-order coefficient remain within the same Gaussian measure class in the dimensional and regularity regime considered here. The corresponding relative covariance perturbation is therefore Hilbert–Schmidt, although it is generally not finite rank.

The high-frequency mechanism can be seen from a first-order expansion. Write

$$A_1 = A_0 + V, \qquad V = \kappa^2(x) - \kappa_0^2, \tag{16}$$

where V denotes multiplication by the leakage perturbation. To first order,

$$T \approx -\left(VA_0^{-1} + A_0^{-1}V\right). \tag{17}$$

The exact equivalence statement is supplied by the operator theorem; (17) gives the high-frequency mechanism. For bounded smooth V, the singular values of $A_0^{-1}V$ inherit the order--2 elliptic decay

$$s_j(A_0^{-1}V) = O\left(j^{-2/d}\right). \tag{18}$$

Consequently,

$$\sum_j s_j(A_0^{-1}V)^2 \sim \sum_j j^{-4/d} \tag{19}$$

is summable for $d < 4$. Thus a lower-order perturbation can affect infinitely many modes while its relative effect decays rapidly enough to remain Hilbert–Schmidt in $d \le 3$.

2.4.3 Spatially varying conductivity

Now let the principal coefficient change,

$$A_1 = \kappa_0^2 I - \nabla\cdot(K_1(x)\nabla). \tag{20}$$

For the variable-coefficient setting of Bolin and Kirchner (2023, Thm. 4.11 and Lemma 4.9), equivalence requires equality of the principal coefficient:

$$K_1(x) = K_0(x). \tag{21}$$

A nontrivial change in $K(x)$ therefore changes the Gaussian measure class. With the driving-noise normalization held fixed, even a constant rescaling $K_1 = cK_0$, $c \neq 1$, acts as an amplitude change and does not preserve equivalence relative to the fixed reference. For the stationary Laplacian case this is seen directly from

$$\left(\kappa^2 - c\Delta\right)^{-2} = c^{-2}\left(\frac{\kappa^2}{c} - \Delta\right)^{-2}, \tag{22}$$

so a constant principal-coefficient rescaling changes both covariance amplitude and range rather than leaving the Gaussian reference unchanged.

The three departures are summarized as

Physical departure	SPDE location	Relative covariance	Model component
Random smooth pumping	Source	Finite rank; in class	Low-rank correction
Spatially varying leakage	Zeroth order	HS; generally infinite rank ($d \leq 3$)	Lower-order operator
Spatially varying conductivity	Principal part	Outside reference class	Principal operator

The variable-coefficient theorem invoked above assumes smooth coefficients and a smooth boundary. The main conductivity lens used below is smooth, so the coefficient regularity is consistent with that setting; the unit-square domain, however, has corners and is not a smooth boundary. We therefore test whether the predicted refinement signature persists on the polygonal domain. A discontinuous disc lens is considered separately as a robustness experiment.

2.5 Refinement behavior of the relative covariance

The Hilbert–Schmidt condition gives an immediate finite-resolution consequence for in-class departures.

Proposition 2.1 (Bounded refinement energy for an in-class departure)**.** *Let $\{P_N\}_{N\geq 1}$ be nested finite-rank orthogonal projections converging strongly to the identity, let $T \in \mathcal{S}_2$, and define*

$$T_N = P_N T P_N, \qquad D_N = \|T_N\|_{\mathrm{HS}}^2. \tag{23}$$

Then

$$D_N \longrightarrow \|T\|_{\mathrm{HS}}^2 < \infty, \qquad \frac{D_N}{N} \longrightarrow 0. \tag{24}$$

The proof is given in Appendix A. Proposition 2.1 formalizes the refinement signature of an in-class departure: its accumulated squared relative covariance approaches a finite limit.

For a principal-coefficient change, Bolin and Kirchner (2023, Lemma 4.9) establish the non-compactness that underlies failure of equivalence. Non-compactness implies that the relative perturbation does not decay in the manner required of a Hilbert–Schmidt operator, but it does not by itself imply a linear growth rate for D_N. The stronger scaling

$$D_N = O(N), \qquad \frac{D_N}{N} \longrightarrow c > 0, \tag{25}$$

follows from a persistent-mode mechanism: if there exist $\rho, \delta > 0$ such that at least ρN eigenvalues satisfy $|t_{j,N}| \geq \delta$, then

$$D_N = \sum_{j=1}^{N} t_{j,N}^2 \geq \rho N \delta^2. \tag{26}$$

For a smooth principal-coefficient change, the following lemma shows that this mechanism is present and identifies the limits.

Lemma 2.2 (A principal-part departure is multiplication plus compact)**.** *Let $\mathcal{D} \subset \mathbb{R}^2$ be bounded with either a smooth boundary or a convex polygonal boundary, let $K \in C^1(\overline{\mathcal{D}})$ with $K \geq k_0 > 0$, and let*

$$A_0 = \kappa^2 I - \Delta, \qquad A_1 = \kappa^2 I - \nabla\cdot(K\nabla), \tag{27}$$

with the same homogeneous Dirichlet conditions and $\Sigma_j = c\, A_j^{-2}$ for a common constant $c > 0$. Then

$$T = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2} - I = M_m + R, \qquad m(x) = K(x)^{-2} - 1, \tag{28}$$

where M_m is multiplication by m and R is compact on $L^2(\mathcal{D})$. Consequently:

(i) $\sigma_{\mathrm{ess}}(T) = m(\overline{\mathcal{D}})$. *In particular T is not compact unless $K \equiv 1$.*

(ii) Let P_N project onto piecewise-constant functions on a partition of $\mathcal{D}$ into N cells of equal measure whose diameters tend to zero, and let $T_N = P_N T P_N$. Then

$$\frac{D_N}{N} \longrightarrow \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} m^2\,dx, \qquad \frac{1}{N}\sum_{i=1}^{N}(T_N)_{ii}^2 \longrightarrow \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} m^2\,dx, \tag{29}$$

so the off-diagonal energy per node vanishes.

(iii) The empirical eigenvalue distribution of T_N converges weakly to the distribution of $m(X)$ with X uniform on $\mathcal{D}$. Hence the persistent-mode condition holds with $\rho = |\{|m| \geq \delta\}|/|\mathcal{D}|$, and

$$\frac{\mathrm{KL}_N}{N} \longrightarrow \frac{1}{2|\mathcal{D}|}\int_{\mathcal{D}} \big[m - \log(1+m)\big]\,dx. \tag{30}$$

The proof is given in Appendix B. Part (i) recovers the non-compactness of Bolin and Kirchner (2023, Lemma 4.9); parts (ii) and (iii) give the extensive scaling (25) with an explicit constant, which equivalence theory alone does not provide. The lemma is stated for a piecewise-constant Galerkin compression; the finite-difference relative covariance used in the numerical study is a consistent approximation of it, and the agreement of the two is checked numerically in Section 5.1. The proof requires $K \in C^1$ and does not cover the discontinuous disc lens of Section 5.5; there the per-node energy also persists ($D_N/N \approx 0.035$), which suggests that the conclusion extends, but this is not proved here.

The same distinction appears in Gaussian relative entropy. If the eigenvalues of $I + T_N$ remain bounded away from zero, then

$$\mathrm{KL}_N(\mu_1 \,\|\, \mu_0) = \frac{1}{2}\sum_{j=1}^{N}\big[t_{j,N} - \log(1+t_{j,N})\big]. \tag{31}$$

For a bounded Hilbert–Schmidt perturbation this sum can converge, whereas a persistent $O(1)$ shift over a positive fraction of modes produces an $O(N)$ KL cost; for a smooth principal-coefficient change, Lemma 2.2(iii) gives its slope. Thus a principal-part mismatch can appear finite at every fixed discretization while becoming increasingly distant from the reference as the mesh is refined.

2.6 What the continuum theory does not determine

The results above compare known population covariances. They assume access to

$$T_N = \Sigma_{0,N}^{-1/2}\Sigma_{1,N}\Sigma_{0,N}^{-1/2} - I_N \tag{32}$$

along a refinement sequence. In the prior-learning problem, however, $\Sigma_{1,N}$ is unknown. At a given resolution, the replicated fields provide only the sample covariance

$$\widehat{\Sigma}_{1,N} = \frac{1}{n}\sum_{i=1}^{n} u^{(i)}u^{(i)\top} \tag{33}$$

for centered data, and hence the empirical relative covariance

$$\widehat{T}_N = \Sigma_{0,N}^{-1/2}\widehat{\Sigma}_{1,N}\Sigma_{0,N}^{-1/2} - I_N. \tag{34}$$

When N/n is not small, the spectrum and norm of $\widehat{T}_N$ are themselves strongly affected by sampling variability. Under the exact reference model, let

$$z_i \overset{\text{iid}}{\sim} \mathcal{N}(0, I_N), \qquad S = \frac{1}{n}\sum_{i=1}^{n} z_i z_i^{\top}. \tag{35}$$

Even though the population relative covariance is zero,

$$\mathbb{E}\|S - I_N\|_F^2 = \frac{N(N+1)}{n}. \tag{36}$$

Thus the naive sample Hilbert–Schmidt norm grows with dimension under the null and cannot be used directly as a refinement diagnostic. The continuum theory therefore identifies *what* must be distinguished, but not *how* to distinguish it from a finite ensemble of fields. The next section develops the sample-based diagnostics and covariance models used to make that distinction.

3 Statistical methodology

Section 2 identifies the continuum distinction to be inferred: a Hilbert–Schmidt relative covariance departure has bounded accumulated refinement energy, whereas a persistent principal-part departure can retain non-vanishing relative covariance energy per resolved degree of freedom. In practice, however, the population covariance is unavailable and the sample covariance is strongly contaminated by finite-ensemble fluctuations. We therefore separate the statistical problem into two tasks: first diagnose the structure of the departure from replicated fields, and then fit a covariance model whose components are capable of representing that structure.

3.1 Diagnosing covariance departure from finite replicates

3.1.1 Estimated reference and whitening

Let

$$\widehat{\Sigma}_{0,N} = \Sigma_{0,N}(\widehat{\sigma}^2, \widehat{\kappa}) \tag{37}$$

denote the stationary reference covariance obtained by maximum likelihood from the replicated fields. We use this fitted stationary model as a *reference for whitening*, not as the final covariance model. If the discrete reference covariance is written in the general form

$$\Sigma_{0,N} = \sigma^2 A_\kappa^{-1} Q_W A_\kappa^{-T}, \tag{38}$$

with Q_W the discrete white-noise covariance, a corresponding whitening map is

$$z^{(j)} = \widehat{\sigma}^{-1} Q_W^{-1/2} A_{\widehat{\kappa}} u^{(j)}, \qquad j = 1, \dots, n. \tag{39}$$

For the finite-difference discretization used in the numerical study, $Q_W = h^{-2} I$ and (39) reduces to $z = h A_{\widehat{\kappa}} u / \widehat{\sigma}$.

The whitened sample covariance is

$$S = \frac{1}{n} \sum_{j=1}^{n} z^{(j)} z^{(j)\top}. \tag{40}$$

If the reference parameters were known, $\mathbb{E}[S] = I + T_N$. Because the same replicated fields are used to estimate $(\widehat{\sigma}^2, \widehat{\kappa})$, the resulting diagnostic is a plug-in statistic rather than an exactly unbiased estimate of the relative covariance. This distinction is relevant when the stationary range parameter is weakly identified or moves toward the boundary. Such range misspecification remains an in-class covariance deformation, however, and therefore cannot by itself generate a nonzero per-node refinement plateau of the type sought below. We assess the finite-sample sensitivity of the diagnostic to reference fitting separately in the numerical study.

3.1.2 Spectral outliers and finite-dimensional source structure

A finite-rank source perturbation produces isolated population eigenvalues in the whitened covariance,

$$I + T = I + \sum_{k=1}^{r} c_k v_k v_k^\top, \qquad c_k > 0. \tag{41}$$

At finite N and n, even the null covariance produces a nontrivial sample spectrum. Let

$$\gamma = \frac{N}{n}. \tag{42}$$

For whitened Gaussian noise, the upper Marchenko–Pastur edge is

$$\lambda_+ = \left(1 + \sqrt{\gamma}\right)^2. \tag{43}$$

We therefore report the spectral-outlier count

$$\widehat{r}_{\mathrm{MP}} = \#\left\{\widehat{s}_k > \lambda_+\right\}, \tag{44}$$

where $\widehat{s}_k$ are the nonzero eigenvalues of S in decreasing order. The count is used as a diagnostic for isolated positive covariance directions; the fitted correction rank is selected independently by cross-validation.

The outlier count alone cannot diagnose a principal-part departure. A spatially distributed operator change can alter many population modes without producing an isolated sample eigenvalue outside the Marchenko–Pastur bulk. This motivates a second statistic that accumulates a local signature over the spatial discretization.

3.1.3 Diagonal refinement statistic h_N

Let

$$d_i = \frac{1}{n} \sum_{j=1}^{n} z_{ij}^2 \tag{45}$$

denote the empirical variance of the whitened field at node i. For a fixed whitening reference,

$$\mathbb{E}[d_i] = 1 + T_{ii}. \tag{46}$$

Under Gaussian sampling,

$$d_i \overset{d}{=} (1 + T_{ii}) \frac{\chi_n^2}{n}, \tag{47}$$

and therefore

$$\mathbb{E}\left[(d_i - 1)^2\right] = T_{ii}^2 + \frac{2}{n}(1 + T_{ii})^2, \tag{48}$$

$$\mathbb{E}[d_i^2] = \frac{n+2}{n}(1 + T_{ii})^2. \tag{49}$$

This gives the bias-corrected statistic

$$\widehat{H}_N = \sum_{i=1}^{N} \left[(d_i - 1)^2 - \frac{2}{n+2} d_i^2 \right], \tag{50}$$

for which, conditional on a fixed whitening reference,

$$\mathbb{E}[\widehat{H}_N] = \sum_{i=1}^{N} T_{ii}^2. \tag{51}$$

We use the normalized quantity

$$\widehat{h}_N = \frac{\widehat{H}_N}{N}. \tag{52}$$

The statistic is motivated by the distinction in Section 2. If T is Hilbert–Schmidt, then for any orthonormal basis

$$\sum_i T_{ii}^2 \le \|T\|_{\mathrm{HS}}^2 < \infty, \tag{53}$$

so the corresponding per-node diagonal energy vanishes under refinement. A principal-coefficient change has a different local high-frequency signature. By Lemma 2.2, the relative covariance of a smooth scalar conductivity perturbation is multiplication by

$$m(x) = K(x)^{-2} - 1 \tag{54}$$

plus a compact remainder. Consequently, under regular refinement (Lemma 2.2(ii)),

$$\frac{1}{N}\sum_{i=1}^{N} T_{ii}^2 \longrightarrow \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} \left[K(x)^{-2} - 1\right]^2 \, dx =: h_\infty, \tag{55}$$

for the piecewise-constant compression of the lemma; for the finite-difference grid basis used here the limit is a prediction checked against the oracle values. For the conductivity lens specified in Section 4, $h_\infty = 0.060732$, and the finite-grid oracle values approach it from above; the sample-based estimate is compared with both in the Results section.

Equation (51) is exact only when the whitening reference is fixed. Estimating the stationary reference from the same replicates introduces plug-in error, and estimating and removing source directions from the same sample introduces an additional dependence. We therefore refer to (50) as a bias-corrected diagnostic rather than an exactly unbiased test after fitting.

3.1.4 Deflating isolated source directions

Mixed populations can contain both finite-dimensional source variability and a distributed principal-part departure. To expose the latter, we optionally remove isolated positive spectral directions before evaluating (50). Let

$$\widehat{V}_r = [\widehat{v}_1, \dots, \widehat{v}_r] \tag{56}$$

contain the sample eigenvectors associated with eigenvalues above the upper Marchenko–Pastur edge. The whitened fields are projected onto the orthogonal complement,

$$Z_\perp = (I - \widehat{V}_r \widehat{V}_r^\top) Z. \tag{57}$$

Projection reduces the marginal variance at node i by the leverage

$$\ell_i = \sum_{k=1}^{r} \widehat{v}_{ik}^2. \tag{58}$$

The deflated nodal variance is therefore renormalized as

$$d_i^{\perp} = \frac{n^{-1} \sum_j z_{ij,\perp}^2}{1 - \ell_i}, \tag{59}$$

and substituted into (50).

In the experiments considered here, random sources add positive covariance directions while the conductivity lens is *stiff*, $K(x) \geq 1$, so its leading relative-covariance contribution is non-positive. Under this sign structure, upper-edge outliers provide a clean way to identify source directions before evaluating the principal-part diagnostic. For a soft lens, $K(x) < 1$, a principal-part change can itself create positive high-frequency variance; the upper-edge count is then not mechanism-specific, although the diagonal refinement statistic h_N remains sign-insensitive. The present deflation rule should therefore be interpreted as a source-removal step for the sign structure studied here, not as a universal classifier of all principal-part perturbations.

Because $\widehat{V}_r$ is estimated from the same sample used to compute $d_i^{\perp}$, the deflated statistic is only approximately bias corrected. A split-sample or cross-fitted implementation can remove this dependence and is used as a robustness check when evaluating the diagnostic.

3.2 Covariance model classes

The diagnostics above characterize two distinct scales of covariance departure: isolated spectral directions associated with finite-dimensional source structure, and persistent local covariance energy associated with principal-part variation. We compare five model families that place flexibility at different levels of the SPDE.

Model	Flexible component	Intended departure	Role
M1	(σ^2, κ)	stationary reference	baseline
M2	stationary modal spectrum	stationary misspecification	control
M3	finite-rank whitened covariance	random sources	correction
M4-κ	spatial zeroth-order field	leakage	lower-order operator
M4-K	spatial principal coefficient	conductivity	principal operator
M5	M4-K + finite-rank correction	mixed source/operator	combined

3.2.1 M1: stationary reference

The stationary model has covariance

$$\Sigma_{\mathrm{M1}} = \sigma^2 A_\kappa^{-1} Q_W A_\kappa^{-T}, \qquad A_\kappa = \kappa^2 I + L. \tag{60}$$

It defines the fitted reference used for whitening and provides the baseline against which all flexible models are compared.

3.2.2 M2: stationary spectral reshaping

M2 retains the eigenvectors of the stationary Laplacian but allows a smooth deformation of its modal variances. If λ_j are the eigenvalues of L,

$$s_j = \frac{\sigma^2}{h^2(\kappa^2 + \lambda_j)^2} \exp[-\delta(\lambda_j)], \qquad \delta(\lambda) = \sum_{q=1}^{Q} \theta_q \psi_q(\log \lambda), \tag{61}$$

where ψ_q are cubic B-spline basis functions. M2 serves as a flexible stationary control: it tests whether reshaping the stationary spectrum alone is sufficient without introducing spatially varying operator coefficients.

3.2.3 M3: finite-rank correction

M3 augments a stationary SPDE base with a finite-rank correction in whitened coordinates,

$$\Sigma_{\mathrm{M3}} = \sigma^2 A_\kappa^{-1} Q_W^{1/2} \left(I + VCV^\top\right) Q_W^{1/2} A_\kappa^{-T}, \tag{62}$$

where

$$V^\top V = I, \qquad C = \mathrm{diag}(c_1, \ldots, c_r), \qquad c_k \geq 0. \tag{63}$$

For a fixed base, the maximum-likelihood correction directions are the leading eigenvectors of the whitened sample covariance, with spike strengths obtained from the corresponding excess variances. The stationary base parameters and the correction are alternated rather than fitted sequentially, because a stationary range parameter can absorb part of the same low-frequency variance that would otherwise appear as a finite-rank source component.

3.2.4 M4-κ: lower-order adaptation

The leakage model allows the zeroth-order coefficient to vary spatially,

$$A_\theta = L + \mathrm{diag}\left\{\kappa^2 \exp[\Phi\theta]\right\}, \tag{64}$$

where the columns of Φ are smooth radial basis functions. Its covariance is

$$\Sigma_{\mathrm{M4}\text{-}\kappa} = \sigma^2 A_\theta^{-1} Q_W A_\theta^{-T}. \tag{65}$$

This model directly represents the in-class lower-order departure considered in Section 2.4.

3.2.5 M4-K: principal-part adaptation

The conductivity model changes the principal coefficient,

$$A_\theta = \kappa^2 I + L_{K_\theta}, \qquad \log K_\theta(x) = \sum_{q=1}^{Q_K} \theta_q \phi_q(x), \tag{66}$$

giving

$$\Sigma_{\mathrm{M4}\text{-}K} = \sigma^2 A_\theta^{-1} Q_W A_\theta^{-T}. \tag{67}$$

M4-K changes the reference operator itself and is therefore the model class capable of representing spatially varying principal-part structure.

3.2.6 M5: operator adaptation with finite-rank correction

The combined model places a finite-rank correction on top of an adapted principal operator,

$$\boxed{\Sigma_{\mathrm{M5}} = \sigma^2 A_\theta^{-1} Q_W^{1/2} \left(I + VCV^\top\right) Q_W^{1/2} A_\theta^{-T}} \tag{68}$$

with

$$A_\theta = \kappa^2 I + L_{K_\theta}. \tag{69}$$

The two components play different structural roles. The operator A_θ represents population variability that changes the principal spatial mechanism, while $VCV^\top$ represents finite-dimensional source variability relative to the adapted operator. M5 therefore implements the rule suggested by Section 2: change the reference operator where the population requires a principal-part change, and represent isolated source variability as a correction around that reference.

3.3 Estimation and model selection

All model parameters are estimated from the training replicates by likelihood or penalized likelihood. The scale parameter σ^2 is profiled analytically whenever possible.

For M1 and M2, stationarity makes the covariance diagonal in the discrete sine-transform basis, so likelihood evaluation reduces to a sum over modal variances. M4-κ and M4-K are fitted by minimizing the penalized negative log likelihood

$$-\ell(\theta, \kappa, \sigma^2) + \lambda_\theta \|\theta\|_2^2, \tag{70}$$

with σ^2 profiled out and gradients computed by automatic differentiation.

For M3 at a fixed rank r, fitting alternates between two conditional updates. Given the stationary base, the fields are whitened and the leading covariance directions and strengths are updated from the whitened sample covariance. Given the correction directions and strengths, (σ^2, κ) are updated by maximizing the Gaussian likelihood. The fitted rank is selected from

$$r \in \{0, \ldots, r_{\max}\} \tag{71}$$

using five-fold held-out log likelihood. The Marchenko–Pastur count $\widehat{r}_{\mathrm{MP}}$ is reported as a diagnostic only; it does not determine the fitted rank.

M5 uses exact block coordinate ascent at the level of the adapted reference. For fixed (κ, θ), the whitened spike block is updated by the PPCA closed form with a free residual variance, jointly updating σ^2, the spike directions, and their strengths. For fixed spike block, (κ, θ) are updated by penalized likelihood with σ^2 profiled in the operator objective. Each accepted block update is non-decreasing in the penalized objective, and cycles continue until the per-cycle tolerance is reached. The rank is again chosen by five-fold cross-validation. The optimization is initialized from a flat conductivity field rather than from the M4-K solution so that the combined model does not inherit operator structure that M4-K may have used to explain source variability.

3.4 Evaluated decision logic

The diagnostic and covariance models are evaluated as complementary parts of the prior-learning problem rather than as a fully automated model-selection algorithm. The spectral and refinement diagnostics have different roles:

$$\text{isolated upper spectral outliers} \quad \longrightarrow \quad \text{finite-dimensional source structure,} \tag{72}$$

whereas

$$\widehat{h}_N \not\longrightarrow 0 \quad \longrightarrow \quad \text{persistent principal-part discrepancy.} \tag{73}$$

This gives the structural logic

Spectral outliers	Deflated $\widehat{h}_N$	Corresponding model structure
present	$\to 0$	finite-rank correction (M3)
absent	non-vanishing	principal-operator adaptation (M4-K)
present	non-vanishing	combined adaptation and correction (M5)

A vanishing principal-part diagnostic does not by itself distinguish a stationary field from a weak lower-order perturbation; the M4-κ comparison is retained for that purpose.

The oracle refinement calculations and their primary predictions were specified before development of the sample-based diagonal refinement statistic $\widehat{h}_N$ and the combined M5 model. The latter were introduced after the initial oracle study to address the finite-sample decision problem and the mixed source/operator case, and are evaluated separately in the subsequent numerical experiments and robustness checks.

4 Numerical study design

The numerical study is designed to separate the structural question posed in Sections 2–3 from the finite-sample fitting problem. All experiments use the same reference operator and differ only in the component through which the population departs from that reference. We first define the discretization and four synthetic populations, then distinguish the oracle refinement calculations from the finite-replicate diagnostic and model-fitting experiments. Results are deferred to Section 5.

4.1 Domain, discretization, and reference process

The common numerical setup is summarized in Table 4.1. The domain is discretized on $n_x \times n_x$ interior nodes with spacing $h = 1/(n_x + 1)$. The operator $-\nabla\cdot(K\nabla u)$ is approximated by a five-point finite-difference stencil; for heterogeneous K, conductance across an edge uses the harmonic mean of the adjacent nodal values. Homogeneous Dirichlet data are imposed outside the interior grid. With nodal white noise represented as $h^{-1}\xi$, $\xi \sim \mathcal{N}(0, I_N)$, a discrete SPDE model has covariance

$$\Sigma = \sigma^2 h^{-2} A^{-1}(I + VCV^\top)A^{-T}, \tag{74}$$

where the spike term is absent unless source variability is included. The assembled finite-difference operators are symmetric in the present study, so $A^{-T} = A^{-1}$ numerically; we retain the transpose notation in discrete covariance formulas to make the covariance construction explicit. This is the finite-difference counterpart of the continuum construction in Section 2.

Table 4.1: Common setup.

Item	Value
Domain	$\Omega = (0,1)^2$, $n_x \times n_x$ interior nodes, $N = n_x^2$, $h = 1/(n_x + 1)$
Boundary condition	homogeneous Dirichlet
Discretisation	finite differences: 5-point stencil for $-\nabla\cdot(K\nabla u)$, harmonic-mean edge conductance $2K_iK_j/(K_i + K_j)$
Reference operator	$A_0 = \kappa_0^2 I + L$, $\kappa_0 = 6$, $K_0 \equiv 1$, $\sigma^2 = 1$
White noise	$u = \sigma h^{-1} A^{-1}\xi$, $\xi \sim \mathcal{N}(0, I_N)$ (nodal variance h^{-2})
Covariance	$\Sigma = \sigma^2 h^{-2} A^{-1}(I + VCV^\top)A^{-T}$
Whitening	$z = hAu/\sigma$
Sampler	sparse LU of A (direct solve)
Mean	zero; fields are not sample-centred
Fields per truth	$n_{\text{train}} = 200$, $n_{\text{test}} = 100$
Seeds	main: $1000 + 10\,\text{id} + n_x$; robustness: $+7919s$, $s = 1, 2, 3$
Sensors (held-out score)	10 fixed points, noise s.d. $0.05\times$ reference RMS
Grids, model fits	$n_x \in \{16, 32\}$
Grids, diagnostics	$n_x \in \{16, 24, 32, 48, 64\}$

4.2 Population departures

The four populations are controlled representatives of source, lower-order, principal-part, and mixed departures. They are summarized in Table 4.2 and visualized in Figure 4.1. A common Gaussian bump

$$b_{c,r}(x) = \exp\left[-\frac{\|x - c\|^2}{2r^2}\right] \tag{75}$$

is used to localize the departures.

Truth A is constructed directly in the reference-whitened coordinates. Three unit-norm Gaussian directions are assigned strengths $c = (90, 60, 40)$, giving

$$\Sigma_A = \sigma^2 h^{-2} A_0^{-1} \left(I + VCV^\top\right) A_0^{-T}. \tag{76}$$

Thus the generative population is exactly of the finite-rank form represented by M3. The smooth nodal directions are discrete approximations of admissible L^2 source profiles; no point-source idealization is used.

Truth B′ changes only the zeroth-order term. The leakage coefficient is reduced locally according to $\kappa(x)^2 = 36 - 30 b_{c_L, r_L}(x)$, producing a smooth dip from 36 in the background to 6 at the lens centre. Truth B instead changes the principal coefficient through

$$K(x) = \exp\left[b_{c_L, r_L}(x)\right], \tag{77}$$

so that K varies smoothly from 1 to e. The main lens therefore satisfies the smooth-coefficient assumption used in the operator classification; the remaining geometric mismatch is the square domain's nonsmooth boundary. Truth C composes the two mechanisms without retuning either one: it uses exactly the B operator together with exactly the A spike directions and strengths.

Table 4.2: The truths. $b_{c,r}(x) = \exp(-\|x - c\|^2/2r^2)$; $v_k = b_{c_k, r_w}/\|b_{c_k, r_w}\|_2$ (nodal norm).

Truth	Operator A	Spikes $VCV^\top$	Parameters
ref	$\kappa_0^2 I + L$	—	$\kappa_0 = 6$
A (wells)	$\kappa_0^2 I + L$	$\sum_k c_k v_k v_k^\top$	centres $(0.25, 0.30)$, $(0.70, 0.25)$, $(0.55, 0.75)$; $r_w = 0.05$; $c = (90, 60, 40)$
B′ (leakage)	$\mathrm{diag}(\kappa(x)^2) + L$	—	$\kappa(x)^2 = \kappa_0^2 + \gamma\, b_{c_L, r_L}$, $\gamma = -30$ (κ^2 from 36 to 6 at the centre)
B (lens)	$\kappa_0^2 I + L_K$	—	$K = \exp(\beta\, b_{c_L, r_L})$, $\beta = 1$: smooth, $K \in [1, e]$
C (wells+lens)	$\kappa_0^2 I + L_K$	as A	as A and B

Lens and leakage: $c_L = (0.5, 0.5)$, $r_L = 0.12$; off-grid robustness lens $c_L = (0.42, 0.57)$.
Robustness of the oracle diagnostic only: disc lens $K = e\, \mathbb{1}_{\|x - c_L\| \le 0.12} + \mathbb{1}_{\text{else}}$; constant $K \equiv 1.5$.

4.3 Replicates and resolution regimes

Each main fit uses $n_{\text{train}} = 200$ independent zero-mean training fields and $n_{\text{test}} = 100$ independent test fields; the realizations are not sample-centred. Model fitting is performed at $n_x \in \{16, 32\}$, while the refinement diagnostic is evaluated at $n_x \in \{16, 24, 32, 48, 64\}$. The latter sequence deliberately increases the aspect ratio $\gamma = N/n_{\text{train}}$ while holding the ensemble size fixed, thereby testing whether the diagnostic remains informative as covariance estimation becomes increasingly high dimensional. Table 4.3 reports the resulting Marchenko–Pastur edge and the number of grid cells resolving each localized feature. In particular, the well width is smaller than one mesh spacing at the coarsest grid, so refinement changes both the statistical aspect ratio and the spatial resolution of the source profiles. Consequently, spike strengths estimated at $n_x = 16$ are not interpreted as directly comparable discretizations of the finer-grid source amplitudes, and the observed decay of source-only refinement statistics begins from an under-resolved spatial representation.

For the sample-based refinement experiment, five independently generated training ensembles of 200 fields are used at each truth and resolution. The main model-comparison data use the deterministic seed rule in Table 4.1; additional seeds are reserved for robustness checks.

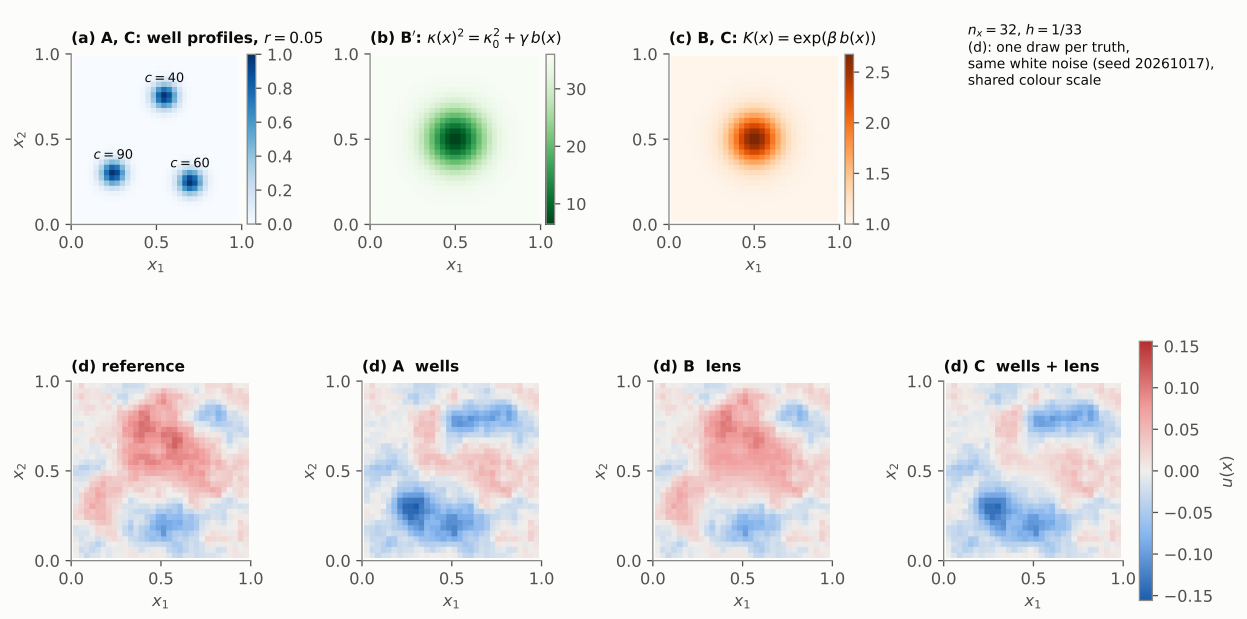


Figure 4.1: Synthetic population construction at $n_x = 32$. The top row shows the three normalized well profiles used in A and C, the spatially varying leakage coefficient used in B′, and the smooth conductivity field used in B and C. The bottom row shows one draw from the reference, A, B, and C generated with the same white-noise realization and displayed on a common colour scale.

Table 4.3: Regimes per grid ($n_{\text{train}} = 200$). $\sqrt{\gamma}$ is the BBP detection threshold for a whitened spike; features are resolved by r/h cells.

n_x	N	$\gamma = N/n_{\text{train}}$	$\sqrt{\gamma}$	MP edge $(1+\sqrt{\gamma})^2$	r_w/h	r_L/h
16	256	1.28	1.131	4.543	0.85	2.04
24	576	2.88	1.697	7.274	1.25	3.00
32	1024	5.12	2.263	10.645	1.65	3.96
48	2304	11.52	3.394	19.308	2.45	5.88
64	4096	20.48	4.525	30.531	3.25	7.80

4.4 Oracle refinement quantities and predictions

Because the synthetic population covariance is known, the first experiment evaluates the population relative covariance directly,

$$T_N = \Sigma_{0,N}^{-1/2} \Sigma_{1,N} \Sigma_{0,N}^{-1/2} - I_N, \tag{78}$$

without sampling error. The oracle calculations record the relative spectrum, the Hilbert–Schmidt energy $D_N = \|T_N\|_F^2$, the normalized energy D_N/N, and the Gaussian divergence $\mathrm{KL}_N(\mu_1 \| \mu_0)$. They also record the diagonal quantity

$$h_N^{\text{oracle}} = \frac{1}{N} \sum_{i=1}^{N} T_{ii}^2, \tag{79}$$

which is the population target of the sample statistic in Section 3.1.3.

For the configured smooth conductivity lens, the local-symbol prediction is

$$h_\infty = \frac{1}{|\Omega|}\int_\Omega \left[K(x)^{-2} - 1\right]^2 \, dx = 0.060732. \tag{80}$$

Midpoint quadrature from 256^2 through 2048^2 points agrees at the reported precision. The exact discrete oracle values for $n_x = 16, 24, 32, 48, 64$ are, respectively,

$$h_N^{\mathrm{oracle}} = (0.06380,\ 0.06363,\ 0.06325,\ 0.06266,\ 0.06228), \tag{81}$$

and approach the continuum prediction $h_\infty = 0.060732$ from above. At the finest configured grid the discrete value remains approximately 2.5% above the limiting value. Accordingly, the sample statistic $\widehat{h}_N$ is compared with h_N^{oracle} at each finite grid, while h_∞ is used only as the asymptotic target. These quantities depend only on the configured truth and reference and are computed before any sample-based fitting.

4.5 Sample diagnostic protocol

At each truth and resolution, M1 is first fitted to the 200 training fields and used as the plug-in whitening reference. The whitened sample covariance is then analyzed at two scales. Positive isolated directions are counted using the unmodified Marchenko–Pastur upper edge

$$\lambda_+ = (1 + \sqrt{N/n_{\mathrm{train}}})^2, \tag{82}$$

with multiplicative edge factor 1.0, i.e. no finite-sample safety margin above the asymptotic edge. As a consequence, an isolated eigenvalue may occasionally cross the threshold even under the reference population; the outlier count is therefore interpreted as a diagnostic rather than a formal level-controlled test. When positive outliers are removed before evaluating the diagonal refinement statistic $\widehat{h}_N$, the nodal leverage correction in (59) uses a denominator floor of 10^{-3}. The finite-sample variance term in (50) is subtracted with the factor $2/(n+2)$ using the actual number of fields in the evaluated ensemble.

The principal diagnostic is the diagonal refinement statistic $\widehat{h}_N = \widehat{H}_N/N$, with and without positive-spike deflation, compared against the oracle target (79). Because both the fitted reference and deflation directions are estimated from data, an additional cross-fitted analysis estimates these quantities on one subset of fields and evaluates $\widehat{h}_N$ on an independent subset. The sign-specific deflation rule is used only for the stiff-lens setting $K \geq 1$ studied here; the diagonal refinement statistic h_N itself is sign insensitive.

4.6 Model-fitting protocol

All covariance models defined in Section 3.2 are fitted to the same training ensembles. Table 4.4 records the parameterizations, initializations, basis sizes, and optimization schemes. M1 is initialized at the reference values and optimized without bounds on κ. M2 uses six cubic B-splines in $\log\lambda$ as a flexible stationary control. M3-joint searches ranks 0 through 8 by five-fold held-out likelihood and alternates between the stationary base and a positive-semidefinite spike correction; in its PCA step the whitened baseline noise variance is fixed at one.

M4-κ and M4-K use 25 Gaussian radial basis functions centred on a 5×5 grid in $\{0.1, 0.3, 0.5, 0.7, 0.9\}^2$, each with width 0.12. Their coefficient penalty is

$$\lambda_\theta \|\theta\|_2^2, \qquad \lambda_\theta = 0.01, \tag{83}$$

added to the total training negative log-likelihood. The value is fixed by hand rather than selected by cross-validation; because the likelihood scales with both the number of fields and the field dimension, this penalty is small relative to the data term. All held-out scores are computed from the unpenalized likelihood.

M5 searches ranks 0 through 5 by the same five-fold criterion but differs from M3 in its spike update. Given the current operator, M5 applies the PPCA closed form with a free residual variance, jointly updating σ^2, the positive spike strengths, and their directions. Given that spike block, the principal-operator coefficients are updated by L-BFGS. The block coordinate-ascent cycle is monotone up to numerical tolerance and stops on a per-cycle objective tolerance. Every M5 fit begins from a flat conductivity field rather than the corresponding M4-K solution.

Table 4.4: Model parameterizations and fitting protocols. Rank is selected by five-fold held-out log likelihood where applicable; numerical tolerances are listed in Table C.1.

Model	Representation	Dimension / rank	Fitting	Initialization / regularization
M1	$\sigma^2 h^{-2}(\kappa^2 + \lambda)^{-2}$ in the DST basis	2 parameters	L-BFGS-B; σ^2, κ joint	$\sigma^2 = 1$, $\kappa = 6$; no penalty
M2	M1 $\times e^{-\delta(\lambda)}$, with six cubic B-splines in $\log \lambda$	8 parameters	L-BFGS-B; joint fit	M1 fit, $\theta = 0$; no penalty
M3-joint	M1 base plus PSD spikes in whitened coordinates	$2 + r(N + 1)$; $r = 0, \ldots, 8$ by CV	Alternating PCA and L-BFGS-B; whitened noise fixed at 1 in PCA block	Two starts: M1-based and $(1, \kappa_0)$; no penalty
M4-κ	$\kappa(x)^2 = \kappa^2 \exp(\Phi\theta)$; 25 Gaussian RBFs	27 parameters	L-BFGS-B with JAX gradients; σ^2 profiled	$\kappa = \hat{\kappa}_{\mathrm{M3}}$, $\theta = 0$; $\lambda_\theta \|\theta\|^2$
M4-K	$\log K = \Phi\theta$; same 25 RBFs	27 parameters	L-BFGS-B with JAX gradients; σ^2 profiled	$\kappa = \hat{\kappa}_{\mathrm{M3}}$, $\theta = 0$; $\lambda_\theta \|\theta\|^2$
M5	M4-K base plus PSD whitened spikes	$27 + r(N + 1)$; $r = 0, \ldots, 5$ by CV	Exact block coordinate ascent: PPCA spike block and L-BFGS operator block	Flat K, $\kappa = \hat{\kappa}_{\mathrm{M3}}$; $\lambda_\theta \|\theta\|^2$

For M3-joint, the M1-based start uses $(\hat{\sigma}^2_{\mathrm{M1}}, \max(\hat{\kappa}_{\mathrm{M1}}, 10^{-3}))$; the second start is $(1, \kappa_0)$. The 10^{-3} value is an initialization floor only. For M4 and M5, $\lambda_\theta = 10^{-2}$ is fixed by hand and added to the total training negative log-likelihood; held-out scores are unpenalized. RBF centres are $\{0.1, 0.3, 0.5, 0.7, 0.9\}^2$ with width 0.12.

4.7 Evaluation metrics

Models are evaluated on the same 100 held-out fields generated independently from each truth. The primary population score is the full-field Gaussian log likelihood. For presentation, each fitted model is paired with the corresponding truth model on the same test fields and reported as a log-score difference in nats per field, so zero is the population-model benchmark.

A second evaluation conditions on ten noisy point observations. Let H select the sensor nodes and

$$y = Hu + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_y^2 I), \tag{84}$$

with σ_y equal to 5% of the reference-field RMS. For each held-out field we record the conditional log predictive density of the full field, $\log p(u_{\mathrm{test}} \mid y)$, and the RMSE of the corresponding Gaussian posterior mean. These sensor-conditioned scores are reported separately from the primary full-field model-comparison tables in Section 5; they test whether a learned prior remains useful when combined with sparse observations rather than evaluating covariance fit alone.

Mechanism recovery is evaluated separately from predictive score. For B and C we record the correlation between the fitted and true $\log K$ fields together with the fitted range parameter $\widehat{\kappa}$. For A and C we record the

selected spike rank, estimated spike strengths, per-direction cosines, and principal-angle cosines between the estimated and true three-dimensional source subspaces. Refinement diagnostics are reported through D_N, D_N/N, KL_N, h_N^{oracle}, and $\widehat{h}_N$.

4.8 Robustness analyses and chronology

The main comparisons are supplemented by targeted robustness analyses. Model rankings are repeated for three additional independently generated training/test seeds. The conductivity fit is repeated after moving the smooth lens centre from $(0.5, 0.5)$ to $(0.42, 0.57)$ so that it no longer coincides with an RBF centre. The oracle refinement calculation is repeated for a discontinuous disc lens and for a constant principal coefficient $K \equiv 1.5$. Additional checks vary the operator penalty λ_θ, use cross-fitted whitening and deflation in the sample diagnostic, vary the replicate count for the weak B′ leakage case, and compare the exact M5 block-coordinate implementation with the earlier legacy update. All main M5 results use the exact block-coordinate implementation; the legacy fit is retained only as a comparison in the robustness analysis.

The sequence of analyses is also recorded explicitly. The physical departures and oracle refinement predictions were specified before the sample-based diagonal refinement statistic $\widehat{h}_N$ and M5 were developed. The latter were introduced subsequently to address the finite-ensemble classification problem and the mixed source/operator population, and are evaluated separately from the oracle calculations.

5 Results: refinement behavior and mechanism attribution

The numerical results are organized around four questions. First, does mesh refinement recover the distinction predicted by the continuum theory when the population covariance is known exactly? Second, can the same distinction be recovered from only 200 replicated fields? Third, what happens when a covariance departure is forced into the wrong structural component? Finally, does the combined model use its operator and finite-rank components selectively when one or both mechanisms are present? The emphasis is therefore on mechanism attribution rather than on predictive ranking alone.

5.1 Oracle refinement separates finite-rank, lower-order, and principal-part departures

Figure 5.1 and Table 5.1 first remove sampling uncertainty and compare the exact relative covariance under refinement. These oracle refinement signatures were specified as predictions before the sample-based diagnostic study. Truth A behaves exactly as a finite-rank perturbation of the reference. Its Hilbert–Schmidt energy and Gaussian divergence are independent of resolution,

$$D_N = 13300, \qquad \mathrm{KL}_N = 88.83, \tag{85}$$

while D_N/N decreases as the number of resolved degrees of freedom grows. The diagonal energy also decreases rapidly with refinement. The coarse-grid values should not be interpreted as an asymptotic rate because the source profiles are under-resolved at $n_x = 16$; the relevant structural observation is that the total relative covariance energy remains bounded as N increases.

The lower-order leakage perturbation B′ has the same qualitative refinement signature without being finite rank. Its total energy approaches a finite value,

$$D_N : 0.613 \longrightarrow 0.586, \tag{86}$$

and its divergence approaches approximately 0.109 nats. Consequently $D_N/N \to 0$, while the oracle diagonal energy falls from 5.8×10^{-4} to 5.0×10^{-6}. This is the numerical signature expected of an infinite-rank but Hilbert–Schmidt departure.

The conductivity lens B behaves differently. Its relative covariance energy is extensive, meaning that D_N grows linearly with N:

$$\frac{D_N}{N} = 0.0646,\ 0.0640,\ 0.0635,\ 0.0628,\ 0.0623 \tag{87}$$

over the five grids. The KL divergence grows nearly exactly linearly with the number of degrees of freedom,

$$\mathrm{KL}_N \approx 0.16 + 0.0315N, \tag{88}$$

with log–log slope 1.001. The local statistic provides an independent view of the same principal-part structure: the exact diagonal energy decreases from 0.06380 to 0.06228 and approaches the symbol prediction $h_\infty = 0.060732$ from above. Thus the smooth conductivity perturbation realizes the persistent-mode refinement behavior discussed in Section 2.5, rather than merely violating equivalence abstractly.

Table 5.1 also checks the two quantitative predictions of Lemma 2.2. First, D_N/N and h_N converge to each other: their gap is 8.3×10^{-4}, 4.0×10^{-4}, 2.3×10^{-4}, 1.1×10^{-4} and 6×10^{-5} over the five grids (from 1.3% to 0.1% of h_N), so the off-diagonal relative covariance energy per node vanishes, as it must for multiplication plus a compact remainder. Second, the lemma predicts the KL slope

$$\frac{1}{2|\mathcal{D}|}\int_{\mathcal{D}} \big[m - \log(1+m)\big]\, dx = 0.030796 \tag{89}$$

for the lens. The exact values $\mathrm{KL}_N/N = 0.0314,\ 0.0318,\ 0.0318,\ 0.0316,\ 0.0315$ lie 2–3% above it, the same relative offset as h_N above h_∞ (2.5% at $n_x = 64$), so the fitted slope 0.0315 is a prediction rather than a fit.

The L^2-Wasserstein distance provides a useful contrast. For B it remains only about 4×10^{-3} across the same refinement sequence even while KL_N grows from 8.0 to 129.0 nats. The fields can therefore remain close in an ordinary L^2 transport metric while their covariance ratio accumulates an extensive high-frequency discrepancy. Truth C superposes the two mechanisms: at the available resolutions its D_N/N is dominated by the large finite-rank source component and decreases with N, while its KL divergence grows from 87.6 to 208.8 nats and its diagonal energy approaches the lens contribution from above. The different summaries therefore separate the finite-dimensional and distributed parts of the mixed departure.

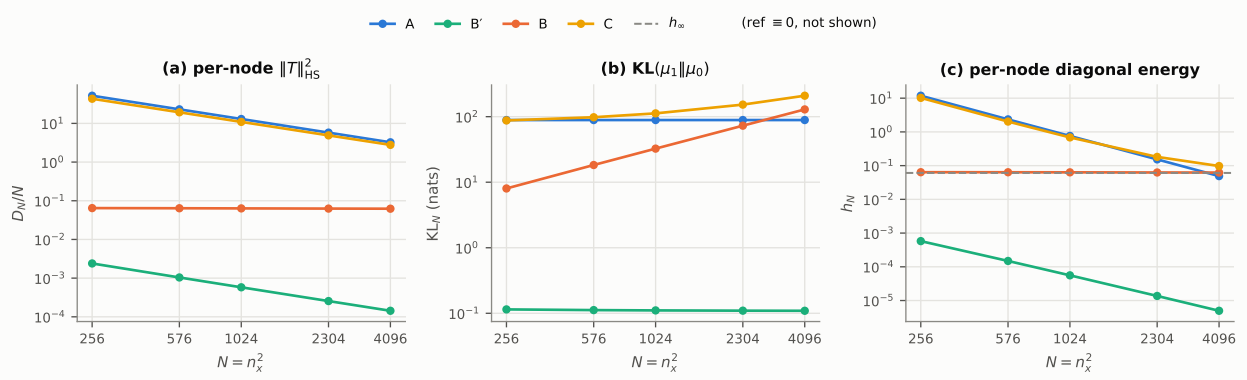


Figure 5.1: Oracle refinement for the four departures (the reference is identically zero). (a) Per-node Hilbert–Schmidt energy $D_N/N = \|T\|^2_{\mathrm{HS}}/N$: constant for the lens (B), decaying for the equivalent departures. (b) $\mathrm{KL}_N(\mu_1\|\mu_0)$: flat for A and B′, while on B it is linear in N, from 8.0 at $N = 256$ to 129 at $N = 4096$; a least-squares line gives $\mathrm{KL}_N \approx 0.16 + 0.0315\,N$ (log–log slope 1.001). C inherits both. (c) Per-node diagonal energy h_N; dashed: the lens limit $h_\infty = 0.0607$.

Table 5.1: Oracle refinement. $D_N = \|T\|^2_{\mathrm{HS}}$, $\mathrm{KL}_N = \mathrm{KL}(\mu_1\|\mu_0)$, $h_N = \frac{1}{N}\sum_i T^2_{ii}$, W_2 in $L^2(\Omega)$ units. Lens limit $h_\infty = 0.060732$.

Truth	n_x	N	D_N	D_N/N	KL_N	h_N	W_2
ref	16	256	0.00	0.0000	0.00	0.00000	0.0000
	24	576	0.00	0.0000	0.00	0.00000	0.0000
	32	1024	0.00	0.0000	0.00	0.00000	0.0000
	48	2304	0.00	0.0000	0.00	0.00000	0.0000
	64	4096	0.00	0.0000	0.00	0.00000	0.0000
A	16	256	13300.00	51.9531	88.83	11.83470	0.0754
	24	576	13300.00	23.0903	88.83	2.35010	0.0748
	32	1024	13300.00	12.9883	88.83	0.75929	0.0746
	48	2304	13300.00	5.7726	88.83	0.15306	0.0745
	64	4096	13300.00	3.2471	88.83	0.04893	0.0744
B′	16	256	0.61	0.0024	0.11	0.00058	0.0040
	24	576	0.60	0.0010	0.11	0.00015	0.0040
	32	1024	0.59	0.0006	0.11	0.00006	0.0040
	48	2304	0.59	0.0003	0.11	0.00001	0.0040
	64	4096	0.59	0.0001	0.11	4.97×10^{-6}	0.0040
B	16	256	16.54	0.0646	8.03	0.06380	0.0043
	24	576	36.88	0.0640	18.30	0.06363	0.0042
	32	1024	65.01	0.0635	32.54	0.06325	0.0042
	48	2304	144.61	0.0628	72.88	0.06266	0.0041
	64	4096	255.33	0.0623	129.01	0.06228	0.0041
C	16	256	11099.86	43.3588	87.61	10.22121	0.0707
	24	576	11149.76	19.3572	98.00	2.02637	0.0701
	32	1024	11193.69	10.9313	112.30	0.68665	0.0699
	48	2304	11285.31	4.8981	152.69	0.18169	0.0697
	64	4096	11400.42	2.7833	208.84	0.09730	0.0697

W_2 at $n_x = 24, 48$ computed for this table with the same routine as the stored values (`run_robust_T.w2`; it reproduces the stored $n_x = 16$ values to 10^{-15}). The reference is identically zero.

5.2 The refinement distinction survives finite sampling

The sample experiment replaces the exact population covariance by 200 replicated fields and whitens them with a fitted stationary M1. The diagonal refinement statistic $\widehat{h}_N$ was developed after the oracle refinement results to ask whether the same distinction could be recovered without access to the population covariance, as recorded in Section 4.8. Figure 5.2 and Table 5.2 show that the two-scale diagnostic retains the oracle distinction despite both finite-sample covariance noise and reference fitting.

Under the reference population, $\widehat{h}_N$ fluctuates around zero at all resolutions. The use of the unmodified Marchenko–Pastur edge occasionally produces one apparent upper outlier: the mean count is 0.2 at $n_x = 32$ and 48. This is consistent with the role assigned to r_{MP} in Section 4.5: it is an outlier diagnostic, not a level-controlled hypothesis test.

Truth A gives a qualitatively different spectral signal. Exactly three upper outliers are detected in every replicate at every resolution. Before deflation the diagonal refinement statistic $\widehat{h}_N$ is dominated by the

localized source directions. After removing those directions, however,

$$\widehat{h}_N^{\mathrm{defl}} = 0.1440,\ 0.0445,\ 0.0173,\ 0.0050,\ 0.0023, \tag{90}$$

and tends toward zero under refinement. This occurs even though the stationary whitening fit drives $\widehat{\kappa}$ toward the boundary, from approximately 1.6×10^{-3} at the coarsest grid to values of order 10^{-4} on the finer grids. The plug-in reference can therefore be badly distorted in its low-frequency range parameter without producing a persistent fine-scale diagonal plateau.

The weak leakage case B′ is also consistent with an in-class departure. It produces no stable outlier count and both raw and deflated $\widehat{h}_N$ fluctuate around zero at the finer grids. In contrast, B remains nearly resolution invariant. The raw statistic is

$$0.0617,\ 0.0624,\ 0.0623,\ 0.0612,\ 0.0612, \tag{91}$$

while the exact finite-grid targets are

$$0.06380,\ 0.06363,\ 0.06325,\ 0.06266,\ 0.06228. \tag{92}$$

For the stiff lens $K \geq 1$, the leading principal-part contribution to the lens block of T is non-positive, so the positive directions deflated on B are occasional finite-sample upper-edge crossings rather than lens signal. After removing them, the estimate lies between 0.0604 and 0.0611, close to the independently specified continuum value $h_\infty = 0.060732$. Thus the persistent principal-part signature is visible from replicated fields without direct access to Σ_1.

The mixed truth C provides the more stringent test because the fitted stationary reference again collapses its range and the raw statistic is dominated by three source directions. Nevertheless, three upper outliers are recovered in every replicate; unlike B, these are the three genuine source directions. After their removal

$$\widehat{h}_N^{\mathrm{defl}} = 0.1702,\ 0.0835,\ 0.0651,\ 0.0585,\ 0.0582. \tag{93}$$

The sequence approaches the same lens-scale plateau seen under B rather than tending to zero as under A. In other words, source deflation exposes the distributed principal-part component that remains underneath the finite-dimensional covariance spikes. Because the deflation directions are estimated from the same fields, the absolute values are interpreted as a bias-corrected diagnostic rather than an exactly unbiased estimator; the refinement pattern is the principal result.

5.3 Wrong-class models fit by reallocating variability to the wrong mechanism

The previous subsections identify the population structures before fitting flexible covariance models. The model comparison asks a different question: when a model class cannot represent the true mechanism directly, which of its available components absorbs the discrepancy? Figure 5.3 and Tables 5.3–5.5 show that these failures are systematic rather than simple losses in predictive score.

For the source-only truth A, the stationary model absorbs the low-frequency effect of the wells by driving its range parameter to the boundary. At $n_x = 16$ and 32,

$$\widehat{\kappa}_{\mathrm{M1}} = 7.2 \times 10^{-5}, \quad 3.0 \times 10^{-5}, \tag{94}$$

instead of the generating value $\kappa_0 = 6$. Giving the model a flexible stationary spectrum improves held-out likelihood but does not repair this mechanism attribution: M2 follows the same collapsed range. Allowing conductivity variation without source spikes leads to a second form of misattribution. This explicit wrong-operator attribution check was added after the oracle refinement results. M4-K places spatial conductivity

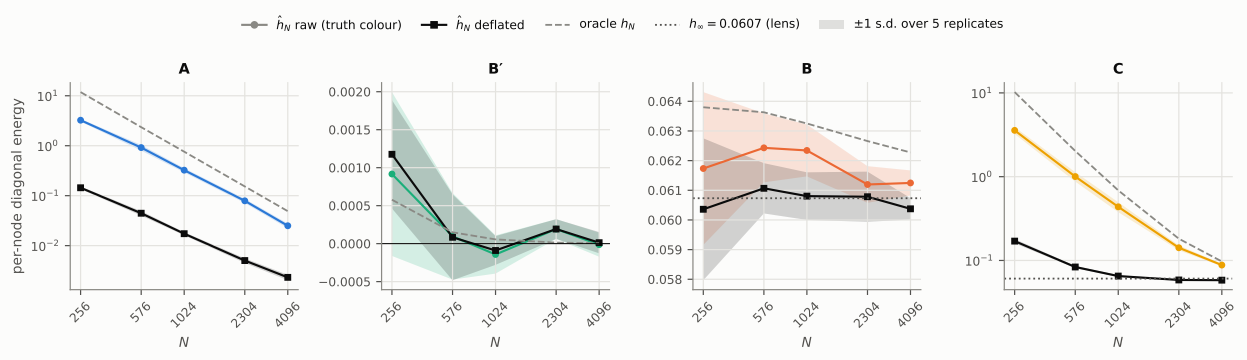


Figure 5.2: Diagonal refinement statistic $\widehat{h}_N$ from 200 fields whitened by a fitted M1, raw (truth colour) and after deflating the eigenvalues above the Marchenko–Pastur edge (black); bands show ±1 standard deviation over five replicates. Dashed curves show the oracle h_N. Dotted curves for B and C show $h_\infty = 0.0607$. On A and B′ the deflated statistic decreases under refinement; on B and C it remains at the lens scale.

Table 5.2: Diagonal refinement statistic from 200 training fields, whitened by a fitted M1 (5 replicates, mean ± s.d.). r_{MP}: eigenvalues above $(1+\sqrt{N/200})^2$ (edge factor 1.0); $\hat{h}_N = \hat{H}/N$ before and after deflating those r_{MP} directions.

Truth	n_x	r_{MP}	$\hat{\kappa}$	$\hat{\sigma}^2$	$\hat{h}_N$ raw	$\hat{h}_N$ deflated
ref	16	0.0 ± 0.0	6.04 ± 0.19	1.003 ± 0.006	0.0002 ± 0.0004	0.0002 ± 0.0004
	24	0.0 ± 0.0	5.98 ± 0.15	1.000 ± 0.005	-0.0003 ± 0.0004	-0.0003 ± 0.0004
	32	0.2 ± 0.4	6.01 ± 0.08	1.000 ± 0.002	0.0001 ± 0.0004	0.0001 ± 0.0004
	48	0.2 ± 0.4	6.10 ± 0.10	1.000 ± 0.002	0.0001 ± 0.0004	0.0002 ± 0.0004
	64	0.0 ± 0.0	6.00 ± 0.07	1.001 ± 0.001	-2.93×10^{-5} ± 0.0002	-2.93×10^{-5} ± 0.0002
A	16	3.0 ± 0.0	0.00163 ± 0.00193	1.437 ± 0.032	3.2405 ± 0.2088	0.1440 ± 0.0092
	24	3.0 ± 0.0	$(1.52 \pm 1.08) \times 10^{-4}$	1.186 ± 0.017	0.9177 ± 0.1201	0.0445 ± 0.0046
	32	3.0 ± 0.0	$(8.34 \pm 6.30) \times 10^{-5}$	1.098 ± 0.007	0.3239 ± 0.0322	0.0173 ± 0.0013
	48	3.0 ± 0.0	$(1.33 \pm 0.83) \times 10^{-4}$	1.041 ± 0.003	0.0789 ± 0.0077	0.0050 ± 0.0005
	64	3.0 ± 0.0	$(2.01 \pm 0.69) \times 10^{-4}$	1.023 ± 0.002	0.0250 ± 0.0028	0.0023 ± 0.0003
B′	16	0.6 ± 0.8	5.45 ± 0.16	0.993 ± 0.009	0.0009 ± 0.0011	0.0012 ± 0.0007
	24	0.2 ± 0.4	5.47 ± 0.12	1.000 ± 0.003	9.65×10^{-5} ± 0.0006	8.36×10^{-5} ± 0.0006
	32	0.2 ± 0.4	5.49 ± 0.15	0.997 ± 0.005	-0.0001 ± 0.0003	-9.02×10^{-5} ± 0.0002
	48	0.0 ± 0.0	5.59 ± 0.11	0.999 ± 0.002	0.0002 ± 0.0001	0.0002 ± 0.0001
	64	0.2 ± 0.4	5.52 ± 0.08	0.999 ± 0.001	-1.36×10^{-5} ± 0.0002	1.27×10^{-5} ± 0.0001
B	16	0.6 ± 0.5	5.96 ± 0.15	0.864 ± 0.005	0.0617 ± 0.0026	0.0604 ± 0.0024
	24	0.8 ± 0.4	6.04 ± 0.17	0.874 ± 0.003	0.0624 ± 0.0012	0.0611 ± 0.0008
	32	1.2 ± 0.4	5.94 ± 0.09	0.871 ± 0.003	0.0623 ± 0.0009	0.0608 ± 0.0008
	48	0.4 ± 0.5	6.01 ± 0.12	0.875 ± 0.003	0.0612 ± 0.0006	0.0608 ± 0.0008
	64	1.0 ± 0.6	6.06 ± 0.09	0.876 ± 0.001	0.0612 ± 0.0004	0.0604 ± 0.0004
C	16	3.0 ± 0.0	$(0.741 \pm 1.247) \times 10^{-3}$	1.254 ± 0.029	3.5717 ± 0.3379	0.1702 ± 0.0100
	24	3.0 ± 0.0	0.00442 ± 0.00359	1.034 ± 0.013	1.0032 ± 0.1295	0.0835 ± 0.0026
	32	3.0 ± 0.0	$(8.16 \pm 5.11) \times 10^{-5}$	0.964 ± 0.010	0.4367 ± 0.0596	0.0651 ± 0.0017
	48	3.0 ± 0.0	0.00331 ± 0.00401	0.915 ± 0.003	0.1418 ± 0.0097	0.0585 ± 0.0004
	64	3.0 ± 0.0	0.00158 ± 0.00291	0.898 ± 0.002	0.0881 ± 0.0030	0.0582 ± 0.0001

Entries are mean ± s.d. over five replicate ensembles. On A and C the M1 $\hat{\kappa}$ collapses to 10^{-5}–10^{-2}; scientific notation is used where needed to retain the spread.

structure around the well locations while again collapsing $\widehat{\kappa}$ to approximately 10^{-4}; its held-out deficit remains 31.1 and 47.4 nats per field at the two resolutions.

Joint source correction resolves both failures. M3-joint selects rank three, returns $\widehat{\kappa} = 6.18$ and 6.21, and improves the held-out scores to -1.76 and -5.78 nats per field, corresponding to deficits of 1.76 and 5.78 nats per field. Fitting the stationary reference first and adding spikes only afterward is substantially worse. M3-nested inherits the collapsed reference and consequently assigns smaller spike strengths to the wells. The generating strengths are 90/60/40; at $n_x = 32$ the nested fit gives 63.1/47.2/29.9, whereas the joint fit gives 101.8/77.3/47.6. Thus the nested fit underestimates all three strengths, while the joint fit overshoots them. The latter is consistent with BBP-type upward sample-eigenvalue inflation in finite-sample spiked covariance models and should not be interpreted as an operator-fit failure. The corresponding source-subspace cosines are also weaker for the nested fit. This demonstrates that a correction can be structurally appropriate yet still be distorted if the reference is estimated independently and allowed to absorb the same variation first.

The converse failure appears under the principal-part truth B. Neither M1, M2, nor M3-joint can represent the conductivity field, and CV correctly selects $r = 0$ for M3. Their held-out deficit grows from about 7.0 nats per field at $n_x = 16$ to 29.0 at $n_x = 32$. Since the number of degrees of freedom increases by a factor of four, the score deficit grows by a factor of 4.14, matching the extensive refinement signature seen in the oracle KL calculation. M4-K, by contrast, remains near the truth at both resolutions, with held-out scores -0.094 and -0.120 nats per field (deficits 0.094 and 0.120) and $\mathrm{corr}(\log \widehat{K}, \log K_{\mathrm{true}})$ increasing from 0.997 to 0.999.

Allowing only a spatially varying lower-order term does not solve this mismatch. M4-κ improves the score only modestly and deforms its coefficient field dramatically: at $n_x = 32$,

$$\frac{\kappa(x)^2}{\widehat{\kappa}^2} \in [1.3, 857]. \tag{95}$$

The optimization audit confirms that this is a converged fit rather than a premature optimizer stop. It is therefore a concrete example of a lower-order model using extreme coefficient variation to imitate a principal-part change that it cannot represent directly.

The weak leakage truth B′ provides the complementary control. M4-κ is the structurally matched model and is the only model with a score indistinguishable from the truth at both resolutions, but all B′ score differences are small relative to their paired standard errors. The experiment therefore supports the structural classification more strongly through refinement than through predictive ranking in this deliberately weak case.

Finally, the mixed truth C exposes the cost of forcing two mechanisms into either component alone. At $n_x = 32$, M3-joint captures the three wells but leaves a held-out score of -34.97 nats per field, i.e. a deficit of 34.97, because the lens remains unmodeled. M4-K instead tries to absorb the source variation into the operator: $\widehat{\kappa}$ collapses to 1.1×10^{-4} and the conductivity-shape correlation reaches only 0.886, despite a peak $\widehat{K}$ close to the true amplitude. Predictive adequacy alone would therefore obscure the structural source of the error; the fitted parameters show which mechanism each wrong-class model is using to compensate.

5.4 The combined model separates source and operator variability

M5 is intended to prevent the compensating failures above by allowing the principal operator and the finite-rank correction to be learned together. The combined architecture was developed after the oracle results exposed the need to represent source and principal-part departures simultaneously, as recorded in Section 4.8. Figure 5.4 shows that the two components are used selectively rather than both being activated indiscriminately. In the diagnostic control at $n_x = 32$, deflation removed 1.2 ± 0.4 upper-edge directions on B, where they are sampling fluctuations, and exactly 3.0 ± 0.0 on C, where they are the three source directions.

For truth A at $n_x = 32$, CV selects $r = 3$, the range returns to $\widehat{\kappa} = 6.08$, and the fitted conductivity remains nearly flat. On this same dataset, M1 alone gives $\widehat{\kappa} = 3.0 \times 10^{-5}$: allowing the source component therefore moves the inferred range by roughly five orders of magnitude back to the generating value $\kappa_0 = 6$. Its held-out

Table 5.3: Held-out log-likelihood relative to the truth (nats per field, 100 test fields; 0 = truth) at $n_x = 16$ and 32. Superscript: CV-selected rank. Bold: best model. ×: gap growth, score at 32 divided by score at 16 (N grows 4×).

	A			B′			B			C		
Model	16	32	×	16	32	×	16	32	×	16	32	×
M1	-53.10	-60.80	1.15	-0.12	-0.14	1.11	-7.00	-28.99	4.14	-60.32	-81.96	1.36
M2	-39.14	-40.84	1.04	-0.11	-0.14	1.28	-6.99	-28.98	4.14	-47.06	-66.39	1.41
M3-joint	**-1.76**$^{(3)}$	**-5.78**$^{(3)}$	3.27	-0.12$^{(0)}$	-0.14$^{(0)}$	1.11	-7.00$^{(0)}$	-28.99$^{(0)}$	4.14	-8.44$^{(3)}$	-34.97$^{(3)}$	4.15
M3-nested	-13.34$^{(3)}$	-10.68$^{(3)}$	0.80	-0.12$^{(0)}$	-0.14$^{(0)}$	1.11	-7.00$^{(0)}$	-28.99$^{(0)}$	4.14	-22.58$^{(3)}$	-39.52$^{(3)}$	1.75
M4-κ	-46.56	-60.80	1.31	**0.01**	**0.00**	0.09	-5.90	-26.54	4.50	-54.84	-78.50	1.43
M4-K	-31.11	-47.42	1.52	-0.11	-0.25	2.30	-0.09	**-0.12**	1.28	-30.40	-42.24	1.39
M5	-1.83$^{(3)}$	-5.86$^{(3)}$	3.20	-0.11$^{(0)}$	-0.25$^{(0)}$	2.30	**-0.09**$^{(0)}$	-0.12$^{(0)}$	1.28	**-1.83**$^{(3)}$	**-6.37**$^{(3)}$	3.49

Paired standard errors (CSV) range from 0.03 to 5.6 nats; largest for the misspecified models on A and C. On B′, all models are within ≤ 0.06 nats standard error, so the apparent 0.01/0.00 advantage of M4-κ is not treated as a significant ranking and the growth ratio is not meaningful.

Table 5.4: Operator recovery. corr = corr($\log \hat{K}, \log K_{\text{true}}$) and peak $\hat{K}$ (true peak $e = 2.718$) for M4-K and M5 on B and C. Range: $\hat{K}$ for M4-K, M5; $\kappa(x)^2/\hat{\kappa}^2$ for M4-κ.

		$n_x = 16$					$n_x = 32$				
Truth	Model	$\hat{\kappa}$	corr	peak $\hat{K}$	range	r_{cv}	$\hat{\kappa}$	corr	peak $\hat{K}$	range	r_{cv}
A	M1	7.17×10^{-5}	—	—	—	—	3.05×10^{-5}	—	—	—	—
	M3-joint	6.18	—	—	—	3	6.21	—	—	—	3
	M4-κ	18.47	—	—	0.000337–0.668	—	4.45×10^{-5}	—	—	0.999–1	—
	M4-K	5.44×10^{-5}	—	—	0.31–1.01	—	9.93×10^{-5}	—	—	0.61–1.06	—
	M5	6.29	—	—	0.999–1.07	3	6.08	—	—	0.943–0.983	3
B′	M1	5.54	—	—	—	—	5.44	—	—	—	—
	M3-joint	5.54	—	—	—	0	5.44	—	—	—	0
	M4-κ	6.20	—	—	0.243–1.36	—	3.24	—	—	0.478–7.12	—
	M4-K	5.66	—	—	0.994–1.1	—	5.51	—	—	1–1.06	—
	M5	5.66	—	—	0.994–1.1	0	5.51	—	—	1–1.06	0
B	M1	6.02	—	—	—	—	6.05	—	—	—	—
	M3-joint	6.02	—	—	—	0	6.05	—	—	—	0
	M4-κ	1.07	—	—	0.0121–126	—	0.48	—	—	1.31–857	—
	M4-K	5.93	0.997	2.46	0.923–2.46	—	5.99	0.999	2.67	0.981–2.67	—
	M5	5.93	0.997	2.46	0.923–2.46	0	5.99	0.999	2.67	0.981–2.67	0
C	M1	3.07×10^{-5}	—	—	—	—	1.77×10^{-4}	—	—	—	—
	M3-joint	5.84	—	—	—	3	6.18	—	—	—	3
	M4-κ	31.88	—	—	0.000195–0.242	—	5.63	—	—	2.04e-09–2.94	—
	M4-K	3.66×10^{-5}	0.613	2.43	0.312–2.43	—	1.11×10^{-4}	0.886	2.75	0.652–2.75	—
	M5	5.77	0.997	2.52	0.956–2.52	3	6.06	0.999	2.61	0.953–2.61	3

corr is left blank on A and B′: their true conductivity is $K \equiv 1$, so $\log K_{\text{true}}$ is constant and the correlation is undefined. M4-κ on B: the extreme range $\kappa(x)^2/\hat{\kappa}^2$ is a converged fit, not an optimiser failure (433 and 597 iterations; final gradient 1.5×10^{-5} and 4.3×10^{-5} per field; restarting L-BFGS from the stored point lowers the objective by $< 10^{-10}$ per field; `data/check_m4kappa_B.json`). It is the zeroth-order model's best imitation of a principal-part change (single start).

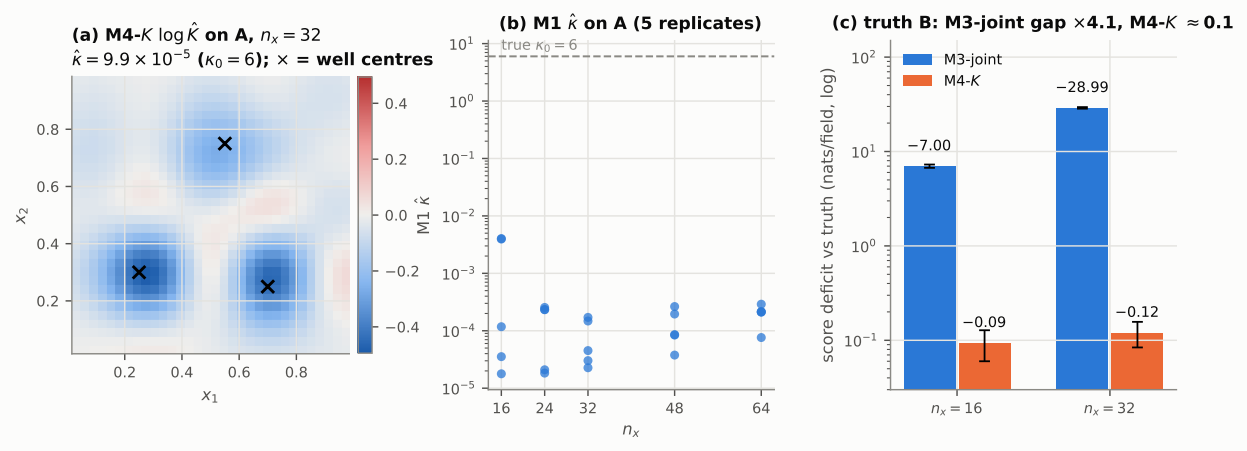


Figure 5.3: Misattribution by wrong-class models. (a) M4-K fitted to truth A at $n_x = 32$ places conductivity structure around the wells and collapses the range ($\widehat{\kappa} = 9.9 \times 10^{-5}$). (b) M1 drives $\widehat{\kappa}$ toward zero on A at every grid. (c) On truth B, the held-out deficit of M3-joint grows from 7.00 to 28.99 nats per field as N grows fourfold, while M4-K remains near −0.1. Error bars are paired standard errors.

Table 5.5: Spike recovery on A and C (true $c = 90/60/40$). $\hat{c}_k = \hat{s}_k - 1$; per-well cosines and principal-angle cosines to the true 3-d well span.

Truth	Model	n_x	r_{MP}	$\hat{c}_{1..3}$	$\cos_{1..3}$	span cosines
A	M3-joint	16	3	99.2/62.5/41.6	0.945/0.945/0.979	0.994/0.990/0.985
		32	4	101.8/77.3/47.6	0.955/0.946/0.938	0.973/0.964/0.941
	M3-nested	16	3	46.4/28.8/19.1	0.924/0.920/0.952	0.977/0.970/0.953
		32	3	63.1/47.2/29.9	0.918/0.905/0.895	0.946/0.931/0.896
	M5	16	—	98.8/61.4/41.5	0.947/0.947/0.979	0.994/0.989/0.985
		32	—	102.8/75.9/47.7	0.958/0.948/0.938	0.973/0.963/0.941
C	M3-joint	16	4	116.4/68.8/41.9	0.987/0.984/0.977	0.992/0.989/0.978
		32	4	112.6/65.6/46.0	0.875/0.859/0.934	0.974/0.955/0.936
	M3-nested	16	3	53.1/31.2/18.8	0.964/0.958/0.951	0.974/0.964/0.951
		32	3	69.1/41.7/28.6	0.853/0.834/0.894	0.941/0.922/0.894
	M5	16	—	109.7/66.0/44.1	0.987/0.984/0.984	0.994/0.990/0.985
		32	—	104.5/62.0/48.5	0.875/0.858/0.945	0.975/0.954/0.946

M3 cosines are from the rank-3 fit (the CV choice is 3 in every case shown). M5 has no r_{MP} (rank by CV only).

score, −5.86 nats per field, is essentially the same as M3-joint (−5.78), showing that adding an operator component does not force spurious conductivity structure when the source correction already explains the population variation. The recovered source subspace is also nearly identical to the M3-joint result: its three principal-angle cosines are 0.973/0.963/0.941.

For B, M5 takes the opposite special case. CV selects $r = 0$, so the finite-rank component switches off. The fit recovers

$$\widehat{\kappa} = 5.99, \qquad \mathrm{corr}(\log \widehat{K}, \log K_{\mathrm{true}}) = 0.999, \qquad \max \widehat{K} = 2.67, \tag{96}$$

against the generating values $\kappa_0 = 6$ and $\max K = e = 2.718$. Its held-out score is numerically identical to M4-K, as expected when no source correction is required.

The mixed truth C requires both components. At $n_x = 32$, M5 selects $r = 3$, recovers

$$\widehat{\kappa} = 6.06, \qquad \mathrm{corr}(\log \widehat{K}, \log K_{\mathrm{true}}) = 0.999, \qquad \max \widehat{K} = 2.61, \tag{97}$$

and retains a three-dimensional source subspace with principal-angle cosines 0.975/0.954/0.946 relative to the true well span. The source strengths are estimated as 104.5/62.0/48.5 compared with the generating values 90/60/40. The corresponding held-out score is −6.37 nats per field (deficit 6.37), compared with scores of −34.97 for M3-joint and −42.24 for M4-K. The gain is therefore not simply due to additional flexibility: the fitted decomposition simultaneously restores the reference range, the conductivity shape, and the finite-dimensional source subspace that each single-component model confounds.

M5 is not intended to replace the lower-order model. On B′, CV selects $r = 0$ and the conductivity field remains close to flat, but M5 does not contain a spatially varying leakage coefficient; M4-κ remains the structurally matched model. This distinction is important for interpreting M5 as a source-plus-principal architecture rather than as a universal covariance approximator.

The exact block-coordinate implementation used for the main results also leaves the scientific conclusion unchanged relative to the earlier legacy alternation. On C at $n_x = 32$, the held-out score changes only from −6.377 to −6.373 nats per field and the conductivity correlation remains 0.9994, while $\widehat{\kappa}$ moves from 5.95 to 6.06. Similar score differences are negligible on A, B′, and B. The optimization correction therefore affects the internal consistency of the fitted decomposition more than the substantive model comparison.

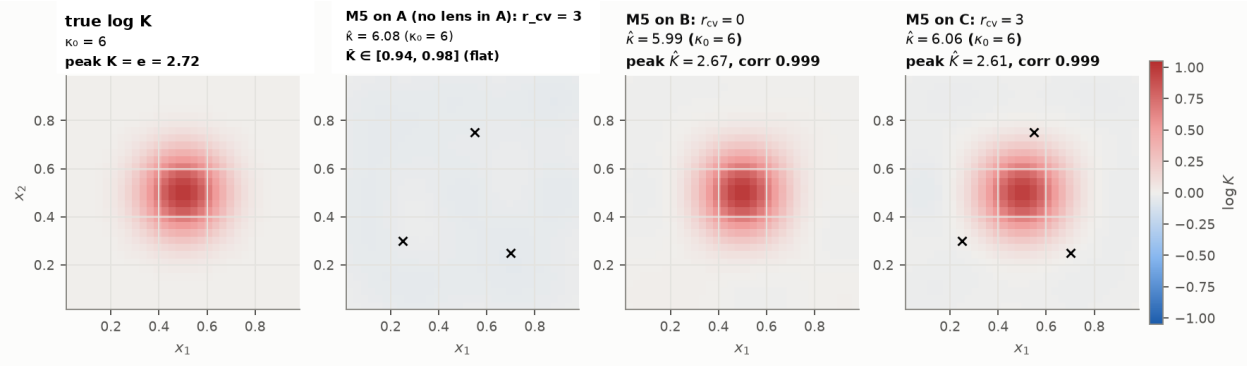


Figure 5.4: M5 selectivity at $n_x = 32$: learned $\log \widehat{K}$ and CV rank ($\kappa_0 = 6$; true peak $K = e$). The first panel shows the true $\log K$ used in truths B and C; truth A has no conductivity lens. A: $r = 3$, $\widehat{\kappa} = 6.08$, nearly flat $\widehat{K}$. B: $r = 0$, $\widehat{\kappa} = 5.99$, peak $\widehat{K} = 2.67$. C: $r = 3$, $\widehat{\kappa} = 6.06$, peak $\widehat{K} = 2.61$.

Table 5.6: M5: exact block coordinate ascent vs the legacy alternation (legacy shown for comparison only). Cycles and final gradient norm $\|\nabla_{(\log\kappa,\theta)}\|$ of the total objective ($n_{\text{train}} = 200$) for the exact fit; the legacy run did not record them.

Truth	n_x	score (leg.)	score (exact)	r_{cv}	$\hat{\kappa}$ (leg.)	$\hat{\kappa}$ (exact)	corr (leg.)	corr (exact)	cycles	$\|\nabla\|$
A	16	-1.83	-1.83	3/3	5.87	6.29	0.0923	0.0563	15	0.301
B′	16	-0.11	-0.11	0/0	5.66	5.66	-0.4548	-0.4548	2	0.002
B	16	-0.09	-0.09	0/0	5.93	5.93	0.9970	0.9970	2	0.001
C	16	-1.87	-1.83	3/3	5.32	5.77	0.9965	0.9974	14	0.259
A	32	-5.86	-5.86	3/3	6.00	6.08	0.0569	0.0511	8	0.301
B′	32	-0.25	-0.25	0/0	5.51	5.51	-0.1949	-0.1949	2	0.015
B	32	-0.12	-0.12	0/0	5.99	5.99	0.9990	0.9990	2	0.010
C	32	-6.38	-6.37	3/3	5.95	6.06	0.9994	0.9994	13	0.304

corr here is against the lens $\log K$ for every truth and is meaningful only for B and C (see Table 5.4).

5.5 Robustness

Across the main seed and three fresh seeds at $n_x = 32$, M5 gives held-out scores -7.04 ± 0.86, -0.10 ± 0.02, and -7.14 ± 0.50 nats per field on A, B, and C and selects ranks 3, 0, and 3, respectively (Table 5.7), so the M5 allocation on C is unchanged and the seed perturbation does not alter the headline h_∞ plateau on B or the $\widehat{\kappa}$ drift on A.

Moving the lens centre off the RBF grid to $(0.42, 0.57)$ increases the $n_x = 32$ M5 deficits to 1.73 nats per field on B and 7.91 on C and reduces the lens correlation to approximately 0.974, but M5 still selects $r = 0$ on B and $r = 3$ on C (Table 5.7), so the allocation headline is unchanged and neither the B plateau nor the A range-drift conclusion is altered by basis placement.

For the discontinuous disc lens, D_N/N remains non-vanishing at 0.0351 and KL_N reaches 104.30 nats by $n_x = 64$ (Table 5.11), so the principal-part refinement signature persists beyond the smooth lens and leaves the smooth-B h_∞ plateau, A range drift, and C allocation headlines unchanged.

For the constant rescaling $K \equiv 1.5$, D_N/N approaches 0.3070 and KL_N reaches 519.22 nats at $n_x = 64$ (Table 5.11), confirming extensive principal-part behavior under a global coefficient change and leaving the three headline conclusions unchanged.

Sweeping λ_θ from 10^{-3} to 10 at $n_x = 32$ leaves B at correlation 0.9990, $r = 0$, and score -0.12 nats per field and C at correlation 0.9994, $r = 3$, and score -6.37 (Table 5.8), so the hand-fixed penalty does not change the B plateau, A range-drift, or C allocation headlines.

Cross-fitting the whitening and deflation on independent 100-field halves gives $\widehat{h}_{64}^{\text{defl}} = 0.0612 \pm 0.0006$ on B, 0.0605 ± 0.0003 on C, and 0.0018 ± 0.0003 on A, compared with in-sample values 0.0604, 0.0582, and 0.0023 (Table 5.9), so the h_∞ plateau and source-versus-principal separation survive removal of same-sample fitting dependence and the three headline conclusions are unchanged.

Increasing the B′ training ensemble from 200 to 800 fields at $n_x = 32$ raises the M4-κ correlation with the true leakage field from 0.800 to 0.874 while M3-joint continues to select $r = 0$ and all held-out score differences remain below 0.15 nats per field (Table 5.10), so the lower-order case becomes more recoverable with replication without changing the B plateau, A range drift, or C allocation headlines.

With ten sensors and 5% noise at $n_x = 32$, M5 on C has conditional log-predictive deficit 6.29 nats per field and posterior-mean RMSE 0.0316 versus truth RMSE 0.0314, compared with deficits 34.89 for M3-joint and 40.38 for M4-K (Table 5.12), so the mechanism allocation remains useful under sparse conditioning and the three headline conclusions are unchanged.

Table 5.7: Robustness: held-out score vs truth (nats per field) over the main seed and three fresh seeds (mean ± s.d.), and the off-grid lens (centre $(0.42, 0.57)$). M5 exact everywhere.

Truth	n_x	seeds	M1	M2	M3-joint	M4-κ	M4-K	M5	M5 $r_{\rm cv}$
A	16	4	-52.45±1.81	-39.32±1.14	-1.74±0.23	-45.61±1.52	-30.40±0.99	-1.81±0.24	3
A	32	4	-60.64±2.00	-40.27±0.89	-6.97±0.86	-58.21±2.46	-47.17±1.30	-7.04±0.86	3
B′	16	4	-0.07±0.03	-0.08±0.02	-0.07±0.03	-0.05±0.04	-0.12±0.02	-0.12±0.02	0
B′	32	4	-0.12±0.03	-0.13±0.01	-0.12±0.03	-0.04±0.05	-0.16±0.06	-0.16±0.06	0
B	16	4	-6.82±0.20	-6.82±0.21	-6.82±0.20	-5.47±0.28	-0.09±0.02	-0.09±0.02	0
B	32	4	-28.47±0.68	-28.47±0.69	-28.47±0.68	-25.98±0.65	-0.10±0.02	-0.10±0.02	0
C	16	4	-57.54±2.04	-45.32±1.30	-8.36±0.16	-51.85±2.03	-29.36±0.96	-1.98±0.30	3
C	32	4	-88.01±4.32	-68.57±1.79	-35.47±0.44	-84.86±5.08	-46.04±2.66	-7.14±0.50	3
B off-grid	16	1	-6.99	-6.99	-6.99	-5.69	-0.37	-0.37	0
C off-grid	16	1	-60.06	-46.83	-8.67	-54.55	-31.29	-2.15	3
B off-grid	32	1	-28.63	-28.62	-28.63	-26.19	-1.73	-1.73	0
C off-grid	32	1	-81.62	-65.75	-34.36	-80.65	-44.67	-7.91	3

M3-nested was not refitted on the robustness seeds (`run_robust_fits.py` skips it); it is omitted here.

Table 5.8: λ_θ sweep (penalty $\lambda_\theta \|\theta\|^2$ on the total negative log-likelihood). True peak $K = e = 2.718$.

Truth	n_x	λ_θ	M4-K corr	M4-K max $\hat{K}$	M4-K score	M5 corr	M5 max $\hat{K}$	M5 $r_{\rm cv}$	M5 score
B	16	0.001	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	0.01	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	0.1	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	1	0.9970	2.45	-0.09	0.9970	2.45	0	-0.09
B	16	10	0.9973	2.40	-0.08	0.9973	2.40	0	-0.08
B	32	0.001	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	0.01	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	0.1	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	1	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	10	0.9990	2.65	-0.12	0.9990	2.65	0	-0.12
C	16	0.001	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	0.01	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	0.1	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	1	0.6134	2.44	-30.39	0.9975	2.51	3	-1.83
C	16	10	0.6159	2.50	-30.38	0.9977	2.44	3	-1.83
C	32	0.001	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	0.01	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	0.1	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	1	0.8864	2.75	-42.23	0.9994	2.61	3	-6.37
C	32	10	0.8868	2.74	-42.22	0.9994	2.60	3	-6.37

Table 5.9: Cross-fitted diagnostic: M1 whitening and deflation directions fitted on 100 fields, $\hat{h}_N$ computed on the other 100, swapped and averaged (5 replicates; same fields as Table 5.2).

Truth	n_x	$\hat{h}_N$ raw	$\hat{h}_N$ deflated	r_{MP}	in-sample deflated
ref	16	0.0008 ± 0.0010	0.0008 ± 0.0010	0.0	0.0002
	24	-0.0004 ± 0.0003	-0.0004 ± 0.0003	0.2	-0.0003
	32	0.0003 ± 0.0005	0.0003 ± 0.0005	0.2	0.0001
	48	0.0001 ± 0.0004	0.0001 ± 0.0004	0.4	0.0002
	64	-4.26×10^{-5} ± 0.0002	-4.04×10^{-5} ± 0.0002	0.2	-2.93×10^{-5}
A	16	3.2633 ± 0.2080	0.1236 ± 0.0089	3.0	0.1440
	24	0.9234 ± 0.1263	0.0320 ± 0.0042	3.0	0.0445
	32	0.3286 ± 0.0312	0.0102 ± 0.0008	3.0	0.0173
	48	0.0787 ± 0.0079	0.0029 ± 0.0007	3.0	0.0050
	64	0.0250 ± 0.0025	0.0018 ± 0.0003	3.0	0.0023
B′	16	0.0008 ± 0.0013	0.0008 ± 0.0013	0.0	0.0012
	24	-7.24×10^{-5} ± 0.0005	-6.43×10^{-5} ± 0.0005	0.1	8.36×10^{-5}
	32	-0.0003 ± 0.0005	-0.0003 ± 0.0005	0.2	-9.02×10^{-5}
	48	0.0001 ± 0.0002	0.0001 ± 0.0002	0.2	0.0002
	64	-6.29×10^{-5} ± 0.0003	-6.04×10^{-5} ± 0.0003	0.3	1.27×10^{-5}
B	16	0.0621 ± 0.0030	0.0621 ± 0.0030	0.6	0.0604
	24	0.0627 ± 0.0012	0.0627 ± 0.0012	0.4	0.0611
	32	0.0621 ± 0.0007	0.0621 ± 0.0007	0.4	0.0608
	48	0.0610 ± 0.0006	0.0610 ± 0.0006	0.5	0.0608
	64	0.0613 ± 0.0006	0.0612 ± 0.0006	0.4	0.0604
C	16	3.5952 ± 0.3750	0.1512 ± 0.0092	3.0	0.1702
	24	1.0091 ± 0.1273	0.0742 ± 0.0020	3.0	0.0835
	32	0.4390 ± 0.0575	0.0614 ± 0.0014	3.0	0.0651
	48	0.1416 ± 0.0096	0.0596 ± 0.0003	3.0	0.0585
	64	0.0880 ± 0.0030	0.0605 ± 0.0003	3.0	0.0582

Table 5.10: Truth B′ with more training fields (nested: the main 200 plus new fields; main test set). Scores in nats per field vs truth; M4-κ corr with the true $\log(\kappa(x)^2/\kappa_0^2)$.

n_x	n_{train}	M1	M1 $\hat{\kappa}$	M3-joint	r_{cv}	r_{MP}	M4-κ	corr
16	200	-0.125	5.54	-0.125	0	0	0.007	0.738
16	400	-0.133	5.64	-0.133	0	0	-0.024	0.856
16	800	-0.138	5.56	-0.138	0	0	-0.042	0.929
32	200	-0.138	5.44	-0.138	0	0	0.001	0.800
32	400	-0.144	5.49	-0.144	0	0	-0.038	0.861
32	800	-0.148	5.52	-0.148	0	0	-0.025	0.874

Table 5.11: Oracle refinement for the smooth lens (B), the disc lens and constant $K \equiv 1.5$.

Truth	n_x	D_N/N	KL_N	W_2
B	16	0.0646	8.03	0.0043
	24	0.0640	18.30	—
	32	0.0635	32.54	0.0042
	48	0.0628	72.88	—
	64	0.0623	129.01	0.0041
B_disk	16	0.0325	5.21	0.0030
	24	0.0403	15.87	—
	32	0.0374	26.73	0.0033
	48	0.0362	59.80	—
	64	0.0351	104.30	0.0032
K_const	16	0.2919	30.39	0.0093
	24	0.2999	70.81	—
	32	0.3032	127.70	0.0090
	48	0.3059	290.71	—
	64	0.3070	519.22	0.0089

Table 5.12: Ten sensors, 5 % noise, $n_x = 32$: conditional log predictive density of the sensor readings relative to the truth (nats per field) and RMSE of the posterior mean field.

Model	A lpd	A RMSE	B′ lpd	B′ RMSE	B lpd	B RMSE	C lpd	C RMSE
truth	0.00	0.0347	0.00	0.0266	0.00	0.0239	0.00	0.0314
M1	-58.89	0.0564	-0.09	0.0266	-28.89	0.0240	-80.28	0.0494
M2	-37.07	0.0562	-0.11	0.0267	-28.89	0.0240	-62.87	0.0496
M3-joint	-5.77	0.0348	-0.09	0.0266	-28.89	0.0240	-34.89	0.0318
M3-nested	-9.87	0.0358	-0.09	0.0266	-28.89	0.0240	-38.65	0.0327
M4-κ	-58.89	0.0564	-0.02	0.0266	-26.07	0.0247	-76.15	0.0497
M4-K	-45.38	0.0563	-0.20	0.0266	-0.12	0.0239	-40.38	0.0501
M5	-5.85	0.0348	-0.20	0.0266	-0.12	0.0239	-6.29	0.0316

6 Discussion and conclusion

The central result is that replicated fields can contain enough information to decide whether a discrepancy from a Gaussian reference should be represented as a correction of that reference or as a change in the covariance-generating operator. In the controlled SPDE study, finite-dimensional source variability and lower-order leakage produced refinement behavior consistent with the reference measure class, whereas a principal-coefficient change produced persistent relative covariance energy and an extensive KL cost. The diagonal refinement statistic $\widehat{h}_N$ converted this continuum distinction into a sample quantity: for the conductivity lens, its finite-grid values tracked the oracle sequence and approached the independently predicted constant $h_\infty = 0.060732$. The same distinction appeared in model fitting. Wrong-class models did not merely lose predictive score; they reassigned variability to physically plausible but incorrect components. Source variability drove a stationary fit from the generating $\kappa_0 = 6$ to $\widehat{\kappa} \approx 3 \times 10^{-5}$, while a lower-order or correction-only model distorted itself to imitate principal-part variation. When both source and operator

variability were available simultaneously, M5 allocated them to separate components and recovered the reference range, conductivity field, and source subspace.

The resulting modeling rule is therefore: *learn the operator where the population demands a change of Gaussian measure class; correct relative to that operator where the remaining discrepancy stays within class*. The rule is structural rather than tied to the aquifer example. A low-rank or Hilbert–Schmidt correction is appropriate when refinement reveals a bounded relative perturbation. When a discrepancy persists across an increasing fraction of resolved modes, adding more correction terms to a fixed reference does not address the underlying mismatch; the reference itself must be changed.

Implications for learned priors and function-space generative models. Methods intended to define probability measures independently of discretization require a reference whose continuum geometry is compatible with the target population. Function-space generative constructions already make absolute-continuity and Cameron–Martin assumptions explicit (Lim et al., 2025; Pidstrigach et al., 2023). The present results add a practical population-learning consequence: if the principal part of the Gaussian reference is wrong, no finite-dimensional correction can repair the measure-class mismatch, even though a fixed mesh can make the corrected covariance appear adequate. The failure becomes visible only as resolution increases. A refinement diagnostic of the type used here can therefore serve as a pre-training check on whether a chosen Gaussian reference is structurally adequate before a more flexible model is learned around it.

Implications for reference-based complexity priors. PC priors quantify complexity as departure from a chosen base model (Simpson et al., 2017; Fuglstad et al., 2019). For a principal-part mismatch, the finite-dimensional Gaussian divergence in this study grows as $\mathrm{KL}_N = O(N)$, so the corresponding KL distance

$$d_N = \sqrt{2\,\mathrm{KL}_N} \tag{98}$$

scales as $O(\sqrt{N})$. The same physical departure therefore appears increasingly complex as the mesh is refined. A discretization-stable notion of complexity in this setting requires either a different distance, as in approaches developed for singular operator changes (Llamazares-Elias et al., 2026), or a new base operator with respect to which the remaining departure is within class. This is an upstream issue to regularization: before asking how strongly to shrink toward a reference, one must ask whether that reference belongs to the right continuum class.

Implications for spatial covariance fitting. The source-only experiment provides a concrete warning about interpreting fitted range parameters. With the wells present, the stationary fit gives $\widehat{\kappa} \approx 3 \times 10^{-5}$ even though the generating value is 6; once source variability is represented explicitly, M5 returns $\widehat{\kappa} = 6.08$ on the same truth. For the range component of the standard Matérn PC construction, the base corresponds to infinite range, or $\kappa = 0$ (Fuglstad et al., 2019). In this example, shrinkage toward that base would therefore reinforce the confounded direction selected by the misspecified stationary model rather than correct it. A near-zero fitted range in the presence of unmodeled sources should consequently be treated as a possible mechanism-attribution warning, not automatically as evidence for truly long-range spatial dependence.

Limitations. Several robustness checks in Section 5.5 reduce implementation-specific concerns: although λ_θ was fixed by hand, sweeping it from 10^{-3} to 10 leaves the B and C M5 ranks and scores unchanged at the reported precision; across four seeds the $n_x = 32$ M5 ranks remain 3/0/3 for A/B/C; moving the lens off the RBF grid lowers the fitted lens correlation to about 0.974 but preserves the B/C rank allocation; a discontinuous disc lens retains non-vanishing D_N/N and a growing KL cost; and the constant rescaling $K \equiv 1.5$ likewise retains extensive principal-part scaling. The remaining limitations are structural. The source-deflation rule was evaluated with a stiff conductivity lens, $K \geq 1$, so although the diagonal refinement

statistic h_N is sign insensitive, the upper-edge source-removal step is not yet a general soft-lens classifier. The study is two-dimensional and employs a square domain whose corners fall outside the smooth-boundary assumptions of the strongest variable-coefficient equivalence theorem. M5 does not include a spatially varying lower-order coefficient, M4 fits use a single initialization, and auxiliary Anderson–Paul-type corrections are applied beyond the regime in which all finite-sample assumptions are rigorously controlled. No real aquifer data are analyzed, and the sample-based diagonal refinement statistic $\widehat{h}_N$ and M5 architecture were developed after the oracle refinement results rather than specified in advance.

Next steps. The immediate extensions follow directly from these boundaries. A soft-lens experiment would separate the sign-insensitive refinement diagnostic from the sign-specific deflation rule, and low-replicate regimes should be studied to determine how much population replication is needed before the refinement classification becomes reliable. Lemma 2.2 should be extended to discontinuous principal coefficients, as the disc-lens computation suggests, to tensor-valued K, and to $d = 3$. A second direction is computational: principal-part hyperparameters generate mutually singular continuum families, so the behavior of centered and noncentered samplers under mesh refinement deserves direct study. The synthetic aquifer study should ultimately be tested against replicated observations from a pumped heterogeneous aquifer. More broadly, the same question arises in mechanics when a learned population prior over constitutive or hardening behavior is transferred to a new experiment: one must determine whether the new population is a correction around the learned reference or whether the operator or constitutive representation defining that reference must change.

The opening question can therefore be answered empirically: whether to correct a Gaussian reference or change its operator is a question that replicated data can answer, and getting that decision wrong can produce a confident, physically plausible, but structurally incorrect model.

A Proof of Proposition 2.1

Proof. If $T \in \mathcal{S}_2$ and $P_N \to I$ strongly, then finite-rank compression converges in Hilbert–Schmidt norm,

$$\|P_N T P_N - T\|_{\mathrm{HS}} \longrightarrow 0. \tag{99}$$

Therefore

$$D_N = \|P_N T P_N\|_{\mathrm{HS}}^2 \longrightarrow \|T\|_{\mathrm{HS}}^2 < \infty. \tag{100}$$

Since the numerator converges to a finite limit while $N \to \infty$,

$$\frac{D_N}{N} \longrightarrow 0. \tag{101}$$

□

B Proof of Lemma 2.2

Proof. Write $\mathcal{K}$ for the compact operators on $L^2(\mathcal{D})$.

Step 1 (regularity). For a smooth or convex polygonal domain and $K \in C^1(\overline{\mathcal{D}})$, $K \geq k_0 > 0$, both A_0^{-1} and A_1^{-1} map $L^2(\mathcal{D})$ boundedly into $H^2(\mathcal{D}) \cap H_0^1(\mathcal{D})$. Hence $A_0 A_1^{-1}$ and $A_1 A_0^{-1}$ extend to bounded operators on $L^2(\mathcal{D})$, and each is the inverse of the other.

Step 2 ($A_1 A_0^{-1} = M_K + \mathcal{K}$). Let $B = A_1 - A_0 = -\nabla \cdot ((K-1)\nabla) = -(K-1)\Delta - \nabla K \cdot \nabla$. Since $-\Delta A_0^{-1} = I - \kappa^2 A_0^{-1}$,

$$B A_0^{-1} = M_{K-1} - \kappa^2 M_{K-1} A_0^{-1} - (\nabla K \cdot \nabla) A_0^{-1}. \tag{102}$$

The operator A_0^{-1} is compact on L^2, and ∇A_0^{-1} maps L^2 boundedly into H^1, which embeds compactly into L^2 (Rellich). Multiplication by the bounded functions $K - 1$ and ∇K preserves compactness. Hence $B A_0^{-1} = M_{K-1} + C_1$ with $C_1 \in \mathcal{K}$, and $A_1 A_0^{-1} = I + B A_0^{-1} = M_K + C_1$.

Step 3 ($A_0A_1^{-1} = M_{1/K} + \mathcal{K}$). Let $G = A_0A_1^{-1}$. From $(M_K + C_1)G = I$,

$$G = M_{1/K} - M_{1/K}C_1G = M_{1/K} + C_2, \qquad C_2 \in \mathcal{K}. \tag{103}$$

Step 4 (the decomposition). Since $\Sigma_0^{-1/2} = c^{-1/2}A_0$ and $\Sigma_1 = c\,A_1^{-2}$, and A_0, A_1 are self-adjoint,

$$T + I = A_0A_1^{-1}A_1^{-1}A_0 = GG^* = (M_{1/K} + C_2)(M_{1/K} + C_2^*) = M_{1/K^2} + C_3, \qquad C_3 \in \mathcal{K}, \tag{104}$$

which is (28) with $R = C_3$. Since G is invertible, $T + I \geq \gamma I$ for some $\gamma > 0$.

(i). By Weyl's theorem, compact perturbations do not change the essential spectrum, so $\sigma_{\text{ess}}(T) = \sigma_{\text{ess}}(M_m)$, the essential range of m, which equals $m(\overline{\mathcal{D}})$ for continuous m. A compact operator has essential spectrum $\{0\}$, so T is compact only if $m \equiv 0$, i.e. $K \equiv 1$.

(ii). In the orthonormal basis of normalized cell indicators, $P_NM_mP_N$ is diagonal with entries the cell averages $\bar{m}_i$, so $\frac{1}{N}\|P_NM_mP_N\|_{\text{HS}}^2 = \frac{1}{N}\sum_i \bar{m}_i^2 \to |\mathcal{D}|^{-1}\int m^2$ by uniform continuity of m. For the remainder, P_NRP_N has rank at most N and singular values $s_j(P_NRP_N) \leq s_j(R)$, so

$$\frac{1}{N}\|P_NRP_N\|_{\text{HS}}^2 \leq \frac{1}{N}\sum_{j=1}^{N} s_j(R)^2 \longrightarrow 0, \tag{105}$$

because $s_j(R) \to 0$ and Cesàro means of a null sequence vanish. The cross term vanishes by Cauchy–Schwarz, giving the first limit in (29). The diagonal entries of T_N are $\bar{m}_i + (P_NRP_N)_{ii}$ with $\frac{1}{N}\sum_i (P_NRP_N)_{ii}^2 \leq \frac{1}{N}\|P_NRP_N\|_{\text{HS}}^2 \to 0$, giving the second limit. Because $P_NM_mP_N$ has no off-diagonal entries in this basis, the difference of the two quantities tends to zero.

(iii). By the Hoffman–Wielandt inequality, with eigenvalues and cell averages both sorted,

$$\frac{1}{N}\sum_{j=1}^{N}\left|t_{j,N} - \bar{m}_{(j)}\right|^2 \leq \frac{1}{N}\|P_NRP_N\|_{\text{HS}}^2 \longrightarrow 0, \tag{106}$$

so the empirical eigenvalue distribution of T_N has the same weak limit as that of the $\bar{m}_i$, namely the law of $m(X)$. The persistent-mode statement follows at every δ with $|\{|m| = \delta\}| = 0$. Finally, all $t_{j,N}$ and $\bar{m}_i$ lie in a compact interval $[\gamma' - 1, \|T\|]$ with $\gamma' = \min(\gamma, \min K^{-2}) > 0$, on which $f(t) = t - \log(1 + t)$ is Lipschitz; hence $\frac{1}{N}\sum_j f(t_{j,N}) - \frac{1}{N}\sum_i f(\bar{m}_i) \to 0$, and the second sum tends to $|\mathcal{D}|^{-1}\int f(m)$. This gives (30). □

C Optimization settings

Table C.1: Optimiser settings (L-BFGS-B, SciPy; gradients by JAX autodiff).

Model	Inner solve	Outer loop	Other
M1, M2	maxiter 1000, maxfun 3000, gtol 10^{-8}, ftol 10^{-13}	—	—
M4-κ, M4-K	maxiter 2000, maxfun 6000, gtol 10^{-8}, ftol 10^{-13}	—	dense Cholesky
M3-joint	maxiter 500, gtol 10^{-10}, ftol 10^{-15}	$\Delta\ell < 10^{-9}$ per field, ≤ 1000	2 starts
M5	maxiter 500, gtol 10^{-8}, ftol 10^{-14}	$\Delta\ell < 10^{-7}n$ per cycle, ≤ 60	monotone check, slack $10^{-8}n$

κ floored at 10^{-3} at initialisation only; no bounds during optimisation.

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